

The Balance Field Equation as a Recursive World Formula

A finite operator model with geometric self feedback and explicit physical bridge requirements

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PREPRINT

Abstract

This preprint specifies the Balance–Field Equation (BFG) as a finite, stratified recursive operator dynamics on unitary equivalence classes. Its declared state is $\Xi=(K,\rho_F,\rho_W,Y,P)$. Spectral selection, canonical polar transport, an endogenous positive metric, formation-dependent loads, an ambient reduced singular-value decomposition, R4 realignment, formation inheritance and a rank-one seed rule define a unique successor class or a named terminal outcome. The complete analysis operator acts on the ambient input space, not only on the selected carrier. This choice has restrictive consequences: R4 is isometric on the selected carrier, its witness mass is exactly one, every nonterminal successor satisfies $0<Y_+<I$, and the ambient dimension cannot increase. Subsequent nonterminal events preserve witness mass. Geometric self feedback is explicit and produces strict damping on a scalar control branch, but neutral kernel and formation directions prevent a universal strict full-state contraction claim. The manuscript gives typed differentials, proof assumptions, event-time bookkeeping and independently reproducible numerical controls. Physical information, seconds, spacetime and gravity, quantum and particle observables, atoms, molecules, biology and experience require additional specified measurement bridges. These transitions are presented in dependency order and remain open where the finite kernel does not determine them. Computed 3D figures demonstrate declared mathematical families; physical-sector illustrations do not constitute empirical validation.

Keywords: Balance–Field Equation; recursive dynamics; polar decomposition; singular-value decomposition; viability; endogenous geometry; quantum bridge; recursive closure; witness persistence; stratified dynamical systems.

1 Introduction

A candidate world formula must do more than compress many concepts into one symbolic expression. It must specify a state space, an evolution law, admissibility conditions, equivalence relations, regime boundaries, and a map from internal variables to measurable quantities. The present construction takes this requirement literally. The BFG is therefore formulated as a recursive operator system whose central object is an orbit of a finite state under a stratified update. The use of polar decomposition and singular-value structure places the mathematical core in a standard matrix-analytic setting [4–8], while viability and Lyapunov concepts provide the appropriate language for constrained persistence [9].

Three co-authored precursor lines are retained as direct BFG provenance. The Level-0 foundation with Thomas W. Loker motivates a pre-informational admissibility layer [1]. The recursive conjugate-closure work with Luke Kenneth Casson Leighton motivates mediated conjugate recursion [2]. The no-supervenience bridge with Alfredo Pereira Jr. motivates the later distinction between functional closure and phenomenal interpretation [3]. No other BFG manuscript is included in the bibliography.

The objective is to present one canonical finite model class, close its mathematical ambiguities, and expose the remaining physical bridge choices. The paper fixes: (i) a reduced-SVD convention and endogenous successor dimension; (ii) a unitary quotient removing common SVD-frame gauge freedom; (iii) Hermiticity and unit compatibility of the formation operator; (iv) normalized witness input and conditional survival masses; (v) named terminal states; and (vi) separate discrete and continuous stability statements.

2 Claim-state hierarchy

Label	Meaning	Use
Definition	Part of the canonical model class	State spaces, update maps, supports, terminal codes
Assumption	Additional constitutive or physical choice	Bridge laws, Lorentzian completion, quantum-state identification
Theorem/Consequence	Derived under stated hypotheses	Gram positivity, scalar damping, gauge covariance
Open bridge	Not fixed by the finite kernel	Physical clock, spacetime, Standard-Model parameters, phenomenality
Empirical hypothesis	Externally testable identification	Quantum bridge A versus B

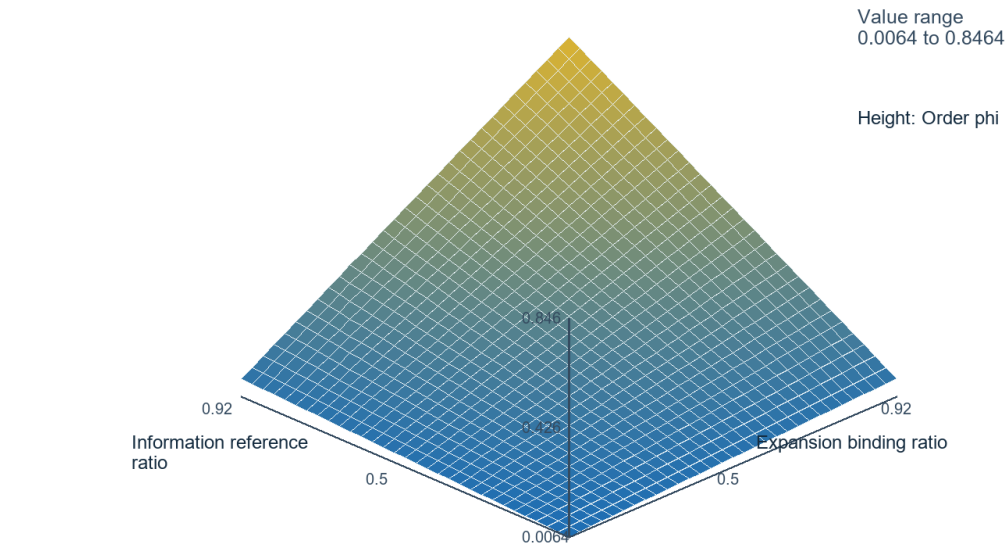
Table 1. Canonical classification and definitions.

The distinction is substantive. A polar factor is a standard matrix object [4,5], but identifying a BFG partial isometry with a physical quantum instrument is an additional physical statement. Likewise, a positive Gram operator is mathematically well defined but is not, by positivity alone, a Lorentzian spacetime metric.

3 Level 0, information, and order

Dimensionless order readout

CONDITIONAL READOUT



$\phi_E = (E_{\text{exp}}/E_{\text{bind}})(I/I_0)$; ratios are independent inputs, not identified Xi data.

Figure 1. Dimensionless order readout. Conditional readout. $\phi_E = (E_{\text{exp}}/E_{\text{bind}})(I/I_0)$; ratios are independent inputs, not identified Ξ data. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

3.1 Level-0 domain

$$\Phi_0 \in X_0 \setminus \{1\}$$

Φ_0 denotes a pre-operational Level-0 domain rather than the later measurable scalar order readout. The co-authored Level-0 construction motivates admissible difference, recurrence, and identity retention before coarse-grained information and physics [1].

3.2 Information

$$I = \log_b |C_\theta(X)| \quad (2)$$

Fix a finite nonempty configuration class with $1 \leq |C_\theta(X)| < \infty$, a threshold or metric θ , logarithm base $b > 1$, and reference $I_0 > 0$. The object X and the counting procedure must be specified independently. This configuration-counting measure differs from Shannon entropy [10]. Neither the count nor its map from Ξ is supplied by the finite kernel.

3.3 Dimensionless order readouts

$$\varphi_N = \frac{N_+}{N_-} \frac{I}{I_0} \quad (3)$$

$$\varphi_E = \frac{E_{\text{exp}}}{E_{\text{bind}}} \frac{I}{I_0} \quad (4)$$

Both readouts are dimensionless for matching numerator and denominator units, with $N_- \rightarrow 0$, $E_{\text{bind}} > 0$ and $I_0 > 0$. They are alternative operational candidates. Φ_0 is a Level-0 object, not a scalar multiplier. No equality between Φ_0 and φ_N or φ_E is assumed.

3.4 Stock and rate

$$q_{\text{info}} = \varphi I \quad (5)$$

$$I_B := dq_{\text{info}}/dt \quad (6)$$

$$dq_{\text{info}}/dt = a_{\text{info}} - \gamma_0 (1 - b_{\text{feedback}}) q_{\text{info}} \quad (7)$$

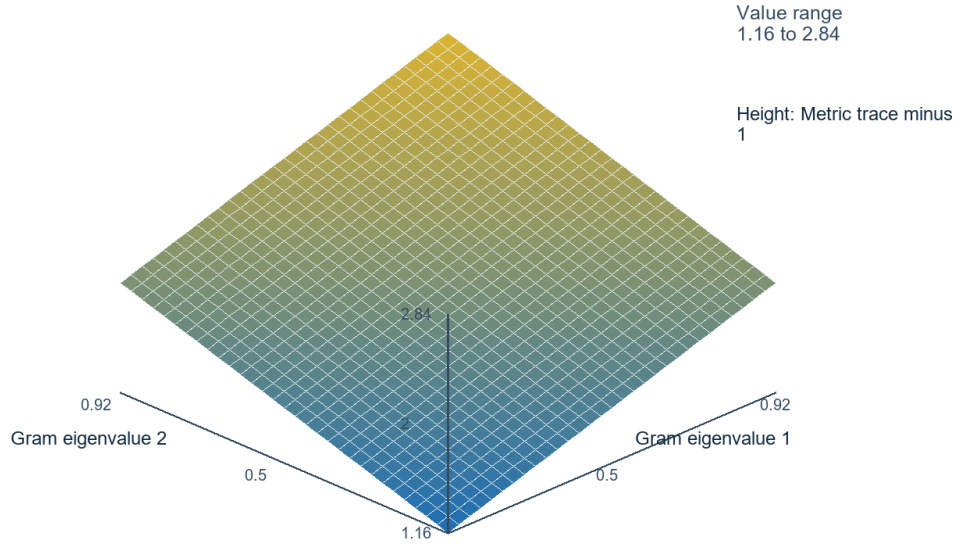
$$q_{\text{info}}^* = \frac{a_{\text{info}}}{\gamma_0 (1 - b_{\text{feedback}})} \quad (8)$$

Here φ denotes one explicitly chosen scalar readout, not Φ_0 . For constant coefficients, $\gamma_0 > 0$ and $b_{\text{feedback}} < 1$, the stated stock is attractive and $I_B^* = 0$. At $b_{\text{feedback}} = 1$, $a_{\text{info}} = 0$ every constant stock is stationary; $a_{\text{info}} \neq 0$ gives linear change. For $b_{\text{feedback}} > 1$ the homogeneous part amplifies. Time-dependent coefficients require the corresponding nonautonomous solution. q_{info} , a_{info} and b_{feedback} are distinct from the selection projector Q , analysis operator A and logarithm base b .

4 Canonical finite state

Positive internal geometry

COMPUTED MATRIX FAMILY



$Y = \text{diag}(y_1, y_2)$; $G = I + Y$; the positive metric is internal, not spacetime.

Figure 2. Positive internal geometry. Computed matrix family. $Y = \text{diag}(y_1, y_2)$; $G = I + Y$; the positive metric is internal, not spacetime. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$\Xi_k = (K_k, \rho_{F,k}, \rho_{W,k}, Y_k, P_k) \quad (9)$$

$$K_k = K_k^\dagger; \rho_{F,k} \geq 0; \rho_{W,k} \geq 0; \text{tr}(\rho_{W,k}) = 1; Y_k = Y_k^\dagger \geq 0; P_k = P_k^\dagger = P_k^2 \quad (10)$$

At event k , all operators act on a finite complex Hilbert space H_k of dimension n_k . K is Hermitian. Formation ρ_F is positive and need not be normalized. Witness ρ_W is positive and normalized at every nonterminal input. Y is a positive semidefinite Gram/load operator and P an orthogonal persistence projector.

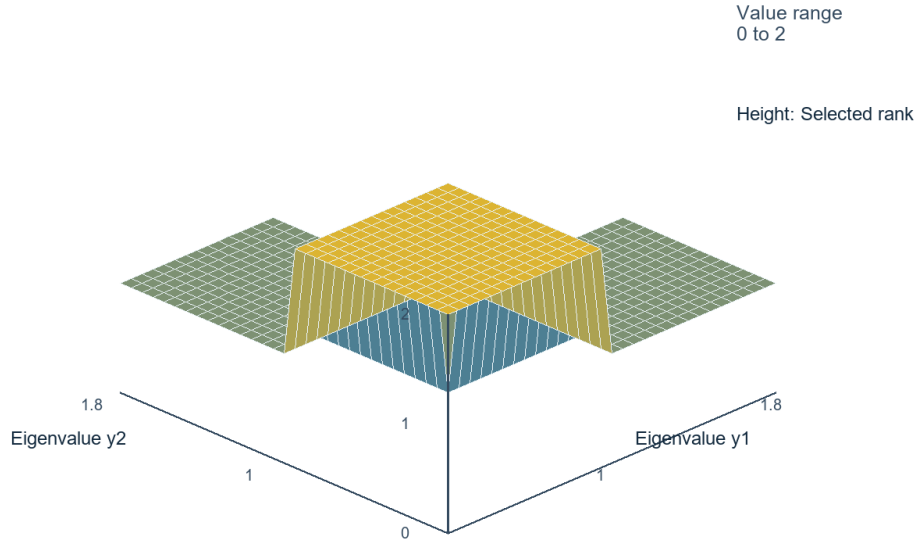
The Level-0 interpretation motivates the model but does not determine its input tuple. A construction $\Xi = J_0(\Phi_0)$ remains an open bridge. The finite state starts only after its five operator components and carrier have been declared.

K and ρ_F share a formation unit E_F in the canonical dimensionally matched model. If a future physical interpretation assigns different units, a fixed positive conversion factor κ_F must enter the seed term; no state-dependent unit conversion is permitted.

5 Spectral selection and first transport

Spectral selection boundary

COMPUTED PROJECTOR



$Q=1_{[0,1)}(Y)$; threshold 1 defines a discontinuous change of selected rank.

Figure 3. Spectral selection boundary. Computed projector. $Q=1_{[0,1)}(Y)$; threshold 1 defines a discontinuous change of selected rank. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$C_k = (I + Y_k)^{-1} \quad (11)$$

$$B_k = Y_k C_k = I - C_k \quad (12)$$

$$G_k = I + Y_k \quad (13)$$

Because Y_k is positive semidefinite, G_k is positive definite. This is the internal Gram geometry.

$$Q_k = 1_{[0,1)}(Y_k) \quad (14)$$

$$X_{\text{sel},k} = Q_k P_k \quad (15)$$

$$T_{\text{sel},k} = \text{polar}(X_{\text{sel},k}) \quad (16)$$

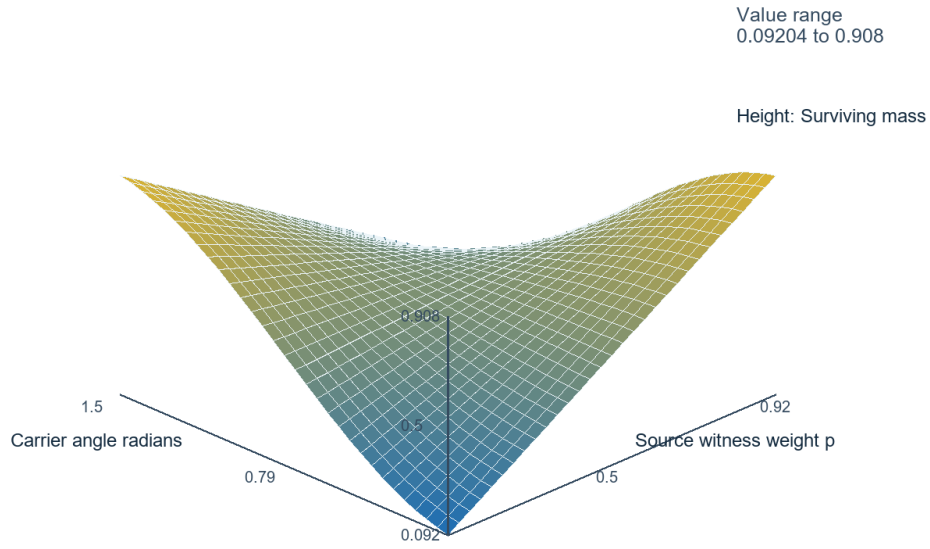
$$S_{\text{sel},k}^- = T_{\text{sel},k}^\dagger T_{\text{sel},k}; P_{\text{sel},k} = S_{\text{sel},k}^+ = T_{\text{sel},k} T_{\text{sel},k}^\dagger \quad (17)$$

The source and target supports are distinct. The polar factor is a partial isometry on the actual supports, consistent with standard polar-decomposition theory [4,5].

6 Formation and witness after selection

Witness mass before normalization

COMPUTED CONTROL FAMILY



$Y=\text{diag}(1/2,2)$, $P=vv^T$ and $W=\text{diag}(p,1-p)$; polar selection has source P .

Figure 4. Witness mass before normalization. Computed control family. $Y=\text{diag}(1/2,2)$, $P=vv^T$ and $W=\text{diag}(p,1-p)$; polar selection has source P . Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$\rho_{F,k}^{\text{sel}} = T_{\text{sel},k} \rho_{F,k} T_{\text{sel},k}^\dagger \quad (18)$$

$$\rho_{\text{un},k}^{\text{sel}} = T_{\text{sel},k} \rho_{W,k} T_{\text{sel},k}^\dagger \quad (19)$$

$$m_{\text{sel},k} = \text{tr}(S_{\text{sel},k} \rho_{W,k}) \quad (20)$$

$$\rho_{W,k}^{\text{sel}} = \rho_{\text{un},k}^{\text{sel}} / m_{\text{sel},k}, \text{ if } m_{\text{sel},k} > 0 \quad (21)$$

Since $\rho_{W,k}$ is normalized and $0 \leq S_{\text{sel}} \leq I$, $0 \leq m_{\text{sel}} \leq 1$. Thus m_{sel} has a genuine conditional-survival interpretation inside the canonical kernel. This is not yet a physical Born probability.

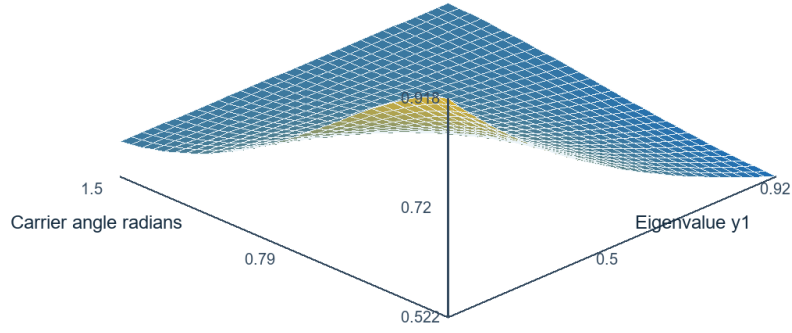
7 Metric projection and endogenous loads

Endogenous formation loads

COMPUTED AMBIENT KERNEL

Value range
0.5216 to 0.9181

Height: C channel load



$Y=\text{diag}(y1,0.72)$; $P=F=W=vv^T$; $v=(\cos \theta, \sin \theta)$; formation unit $E_F=1$.

Figure 5. Endogenous formation loads. Computed ambient kernel. $Y=\text{diag}(y1,0.72)$; $P=F=W=vv^T$; $v=(\cos \theta, \sin \theta)$; formation unit $E_F=1$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$Z_k + Z_k = I; Z_k Z_k^\dagger = P_{\text{sel},k} \quad (22)$$

$$P_{G,k} = Z_k (Z_k + G_k Z_k)^{-1} Z_k + G_k \quad (23)$$

$$\Lambda_{C,k} = \text{tr}(G_k P_{G,k} C_k \rho_{F,k}^{\text{sel}} C_k P_{G,k}^\dagger) \quad (24)$$

$$\Lambda_{B,k} = \text{tr}(G_k P_{G,k} B_k \rho_{F,k}^{\text{sel}} B_k P_{G,k}^\dagger) \quad (25)$$

$$\alpha_k = \frac{\Lambda_{B,k}}{\Lambda_{C,k} + \Lambda_{B,k}} \quad (26)$$

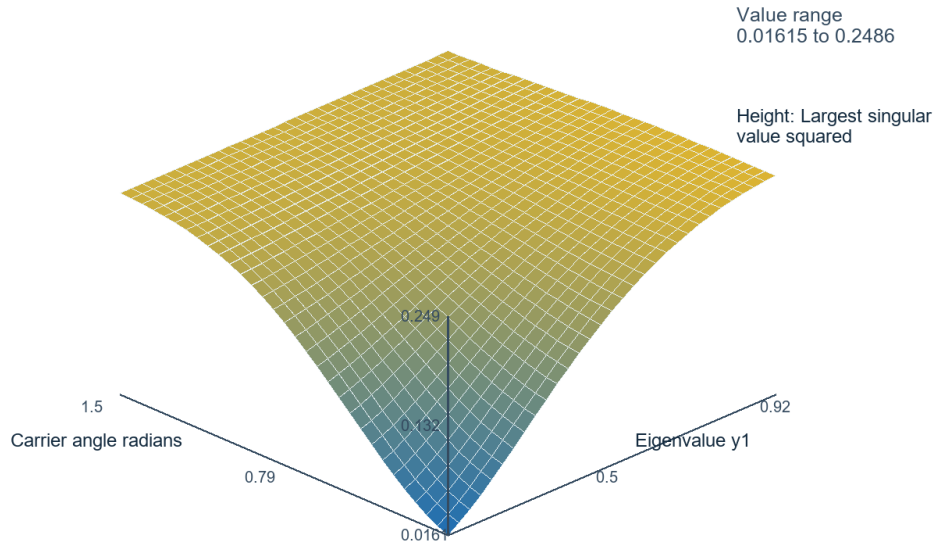
$$\beta_k = \frac{\Lambda_{C,k}}{\Lambda_{C,k} + \Lambda_{B,k}} \quad (27)$$

On the dual-load branch $\Lambda_C > 0$ and $\Lambda_B > 0$, α and β lie in $(0,1)$ and sum to one.

8 Analysis operator, reduced SVD, and endogenous dimension

Full ambient analysis

COMPUTED AMBIENT KERNEL



A:H to H direct sum H; no restriction to the selected rank-one carrier.

Figure 6. Full ambient analysis. Computed ambient kernel. A:H to H direct sum H; no restriction to the selected rank-one carrier. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$A_k = \begin{bmatrix} \sqrt{\alpha_k} P_{G,k} C_k \\ \sqrt{\beta_k} P_{G,k} B_k \end{bmatrix} \quad (28)$$

Let $r_k = \text{rank}(A_k) \geq 1$. The canonical convention uses only the reduced rank- r_k SVD:

$$A_k = U_k \Sigma_k V_k^\dagger \quad (29)$$

$$U_k : C^{r_k} \rightarrow H_k \oplus H_k; V_k : C^{r_k} \rightarrow H_k; \Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_{r_k}), \sigma_i > 0 \quad (30)$$

$$H_{k+1} := C^{r_k}; n_{k+1} = r_k \quad (31)$$

The analysis domain is the full ambient H_k . No restriction $A_k Z_k$ is made. Its codomain is $H_k \oplus H_k$; U and V are column isometries. The successor dimension satisfies $\text{rank}(P_{\text{sel}}) \leq r_k \leq n_k$. It can exceed the selected-carrier dimension, but cannot exceed the ambient input dimension. Null singular directions are omitted. Restricting the domain would define a different kernel and can change the seed complement.

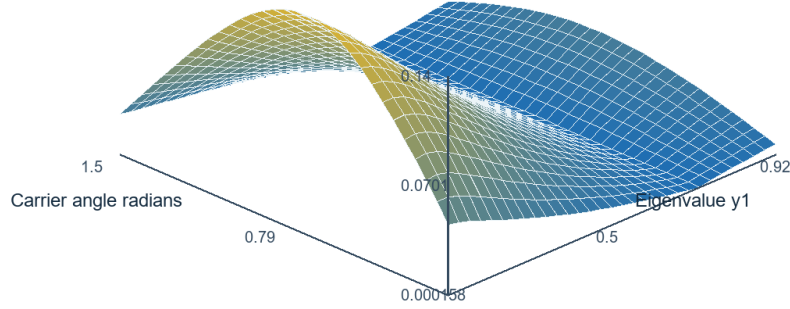
9 SVD gauge and quotient-state dynamics

Reduced singular spectrum

COMPUTED AMBIENT KERNEL

Value range
0.000158 to 0.1401

Height: Second singular
value



The second singular value can vanish on a rank boundary; an arbitrary full-SVD padding is excluded.

Figure 7. Reduced singular spectrum. Computed ambient kernel. The second singular value can vanish on a rank boundary; an arbitrary full-SVD padding is excluded. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

For a simple positive singular value, a common phase remains free. For an exactly degenerate singular block, a unitary basis rotation inside that block remains free. These freedoms are treated as coordinate gauge rather than physical multiplicity.

$$\backslash(K, \rho_F, \rho_W, Y, P) \sim \backslash(W^\dagger K W, W^\dagger \rho_F W, W^\dagger \rho_W W, W^\dagger Y W, W^\dagger P W) \backslash, W \text{ unitary} \quad \backslash(32)$$

For a block-unitary W satisfying $W \Sigma = \Sigma W$:

$$U' = U W ; V' = V W ; \Sigma' = W^\dagger \Sigma W = \Sigma \quad \backslash(33)$$

$$K_+' = W^\dagger K_+ W ; Y_+' = W^\dagger Y_+ W \quad \backslash(34)$$

The R_4 overlap transforms as $X_{R_4}' = W^\dagger X_{R_4}$. Unitary covariance of the polar decomposition gives $T_4' = W^\dagger T_4$, and hence all target-space successor quantities transform by the same conjugation. The mathematical successor is therefore the equivalence class $[\Xi_{k+1}]$, not a privileged matrix representation.

Input covariance is separate from SVD-frame covariance. For a unitary L on H , simultaneous input conjugation gives $A' = (L^\dagger \oplus L^\dagger) A L$. Compatible SVD frames are $U' = (L^\dagger \oplus L^\dagger) U$ and $V' = L^\dagger V$. K_+ and Y_+ then agree, and R_4 transforms consistently. This proves independence of the ambient input basis as well as of the allowed target SVD gauge.

$$U_{\text{bar}}_{\text{BFG}} : [\Xi_k] \rightarrow [\Xi_{k+1}] \quad \backslash(35)$$

This quotient construction solves the definition-level phase/basis ambiguity. It does not claim differentiability of individually sorted singular vectors at eigenvalue splitting. Kato and Davis–Kahan theory instead support the use of invariant subspaces and separated spectral clusters [6,7].

For local differential work on a simple singular-value stratum, with $M = U^\dagger (\delta A) V$:

$$\delta \sigma_i = \text{Re } M_{ii} \quad (36)$$

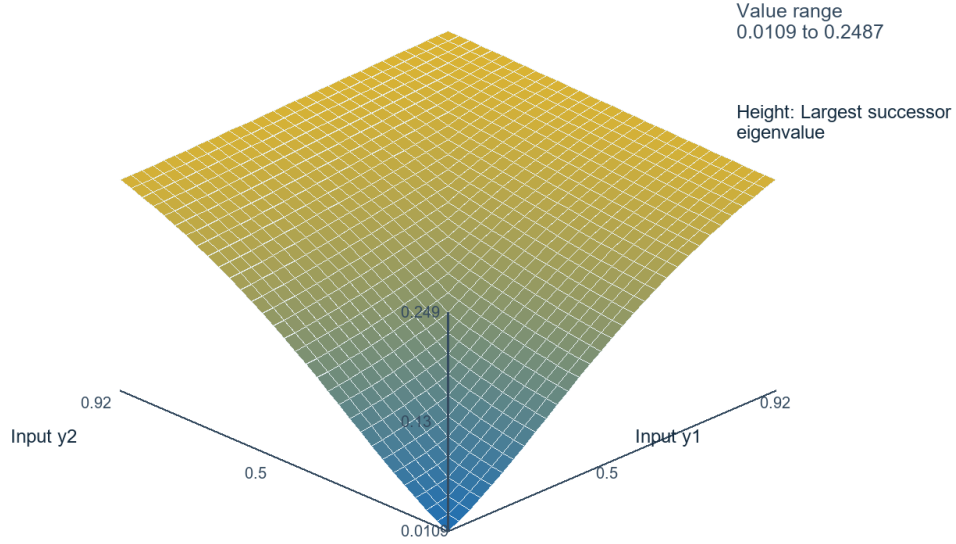
$$(\Omega_U)_{ii} - (\Omega_V)_{ii} = i \text{Im}(M_{ii}) / \sigma_i \quad (37)$$

Only the common phase is free; setting both diagonal gauge connections to zero is generally impossible.

10 Successor geometry and kernel operator

Successor Gram bound

COMPUTED AMBIENT KERNEL



$Y = \text{diag}(y1, y2)$, $P = F = I$; coupled global formation loads determine alpha and beta.

Figure 8. Successor Gram bound. Computed ambient kernel. $Y = \text{diag}(y1, y2)$, $P = F = I$; coupled global formation loads determine α and β . Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$Y_{k+1} = \Sigma_k^2 \quad (38)$$

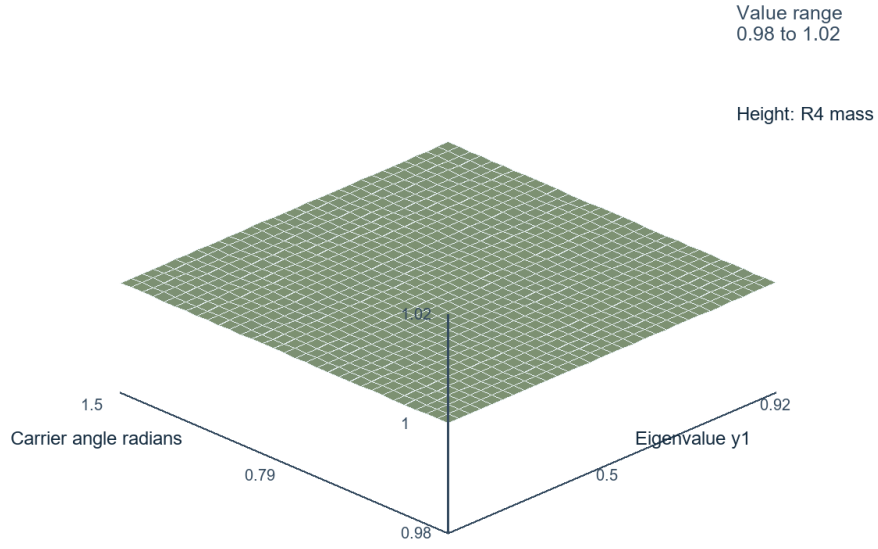
$$K_{k+1} = U_k \dagger (K_k \oplus K_k) U_k \quad (39)$$

Theorem 1. On the positive dual-load branch, the reduced successor satisfies $0 < Y_{k+1} < I_{r_k}$. In particular $G_{k+1} > I$. A proof of this stronger bound, including its limits for full-state stability, follows in Section 17. The input class permits $Y \geq 0$; this is a declared model class, not an assertion that every precursor used the same domain.

11 R4 transport

R4 preserves selected witness mass

COMPUTED AMBIENT KERNEL



$m_4=1$ to numerical precision; the exact proof uses $\text{Ran}(P_{\text{sel}})$ contained in $\text{Ran}(A^\dagger)$.

Figure 9. R4 preserves selected witness mass. Computed ambient kernel. $m_4=1$ to numerical precision; the exact proof uses $\text{Ran}(P_{\text{sel}})$ contained in $\text{Ran}(A^\dagger)$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$X_{R4,k} = V_k \dagger P_{\text{sel},k} \quad (40)$$

$$T_{4,k} = \text{polar}(X_{R4,k}) \quad (41)$$

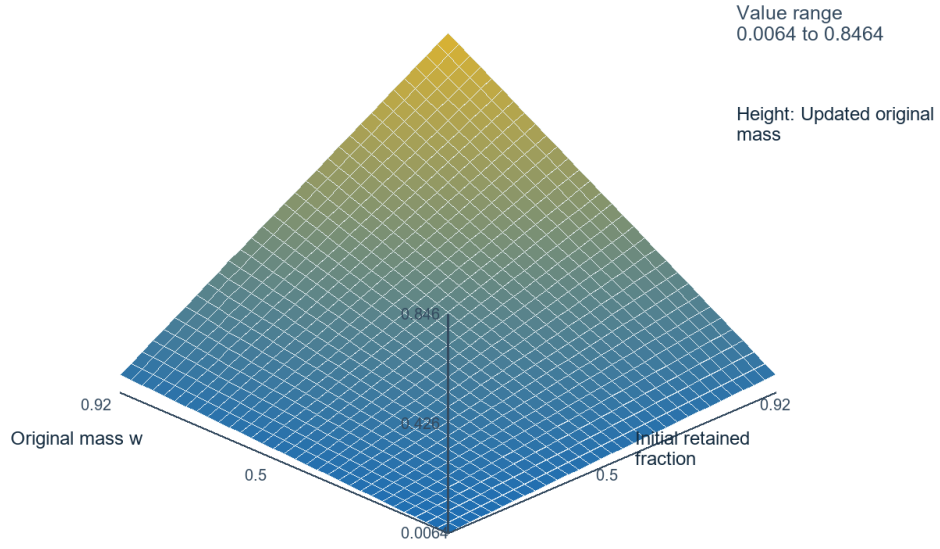
$$S_{4,k}^- = T_{4,k} \dagger T_{4,k}; S_{4,k}^+ = T_{4,k} T_{4,k} \dagger \quad (42)$$

For this ambient kernel, X_{R4} is already a partial isometry: $T_4 = V \dagger P_{\text{sel}}$, $S_4^- = P_{\text{sel}}$ and $S_4^+ = V \dagger P_{\text{sel}} V$. The positive C-channel puts the entire selected carrier in $\text{Ran}(A^\dagger)$. R4 therefore realigns that carrier without a second loss. General rank-loss sensitivity of polar maps remains relevant to selection and to other declared models.

12 Second witness mass and formation inheritance

Original mass and conditioned trace

COMPUTED BOOKKEEPING



$w_{\text{next}} = w * m_{\text{sel}}$; conditioned witness trace remains one and cannot encode this product.

Figure 10. Original mass and conditioned trace. Computed bookkeeping. $w_{\text{next}} = w * m_{\text{sel}}$; conditioned witness trace remains one and cannot encode this product. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$m_{4,k} = \text{tr}(S_{4,k} \rho_{W,k}^{\text{sel}}) \quad (43)$$

$$\rho_{W,k+1} = T_{4,k} \rho_{W,k}^{\text{sel}} T_{4,k}^\dagger / m_{4,k}, \text{ if } m_{4,k} > 0 \quad (44)$$

$$m_{\text{tot},k} = m_{\text{sel},k} m_{4,k} \quad (45)$$

$$\rho_{F,k}^{\text{inh}} = T_{4,k} \rho_{F,k}^{\text{sel}} T_{4,k}^\dagger \quad (46)$$

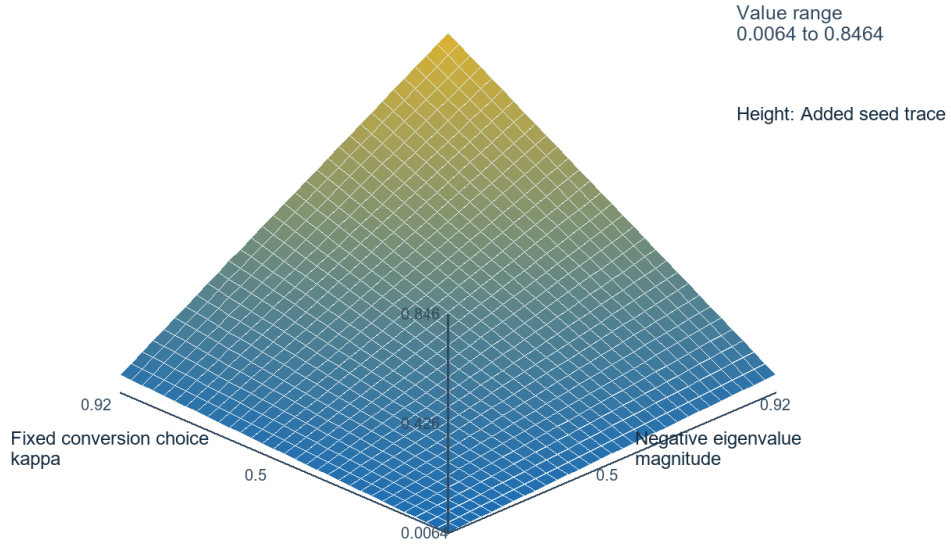
In the present kernel $m_4=1$ exactly, hence $m_{\text{tot}}=m_{\text{sel}}$. The normalized successor does not contain the historical mass. Record $w_{k+1} = w_k m_{\text{sel},k}$, $w_0=1$, as a separate trajectory output. It is not a sixth component silently inserted into Ξ . Directed transport history, when needed, is also recorded separately. Inherited formation satisfies $\text{tr}(\rho_F^{\text{inh}}) = \text{tr}(\rho_F^{\text{sel}})$.

Historical-mass nonidentifiability from the successor: take $Y=\text{diag}(1/2,2)$, $P=I$, $K=-I$, $F=I$ and $W=\text{diag}(p,1-p)$, $0 < p \leq 1$. Selection gives $m_{\text{sel}}=p$, but every p yields the same one-dimensional normalized successor. Therefore Ξ_+ alone cannot recover w or the initial surviving mass.

13 Seed rule

Seed formation on a negative complement

COMPUTED SPECTRAL CONTROL



A one-dimensional complement with $\lambda = -x$ gives seed trace $\kappa \cdot x$; each κ is a separate fixed-unit model.

Figure 11. Seed formation on a negative complement. Computed spectral control. A one-dimensional complement with $\lambda = -x$ gives seed trace $\kappa \cdot x$; each κ is a separate fixed-unit model. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$E_k = I_{r_k} - S_{4,k}^+ \quad (47)$$

If $E_k = 0$, no seed is added. If $E_k \neq 0$, choose an orthonormal frame Z_E for $\text{Ran}(E_k)$ and define:

$$H_{E,k} = Z_{E,k}^\dagger K_{k+1} Z_{E,k} \quad (48)$$

Because K is Hermitian, K_{k+1} and H_E are Hermitian, so the seed spectrum is real.

If H_E has a simple isolated negative minimum $\lambda_k < 0$ with rank-one projector π_k :

$$\Pi_{\text{seed},k} = Z_{E,k} \pi_k Z_{E,k}^\dagger \quad (49)$$

$$\rho_{\text{seed},k} = -\kappa_F \lambda_k \Pi_{\text{seed},k} \quad (50)$$

$$\rho_{F,k+1} = \rho_{F,k}^{\text{inh}} + \rho_{\text{seed},k} \quad (51)$$

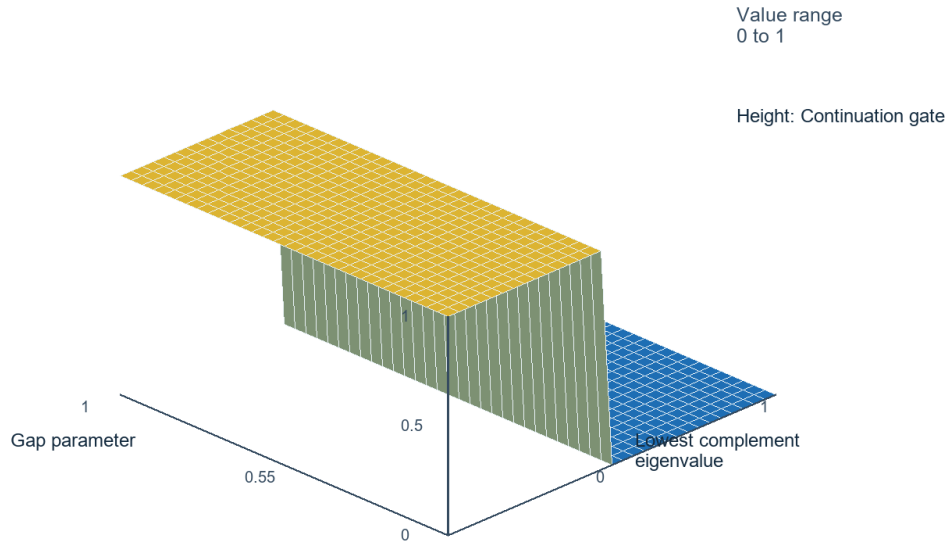
$$P_{k+1} = S_{4,k}^+ + \Pi_{\text{seed},k} \quad (52)$$

For the dimensionally matched canonical class $\kappa_F = 1$. Orthogonality $\Pi_{\text{seed}} \leq E$ implies $S_4^+ \Pi_{\text{seed}} = 0$ and $P_{k+1}^2 = P_{k+1}$.

14 Total stratified update and terminal codes

Terminal boundary of the seed rule

COMPUTED RULE



For a nonempty complement, $\lambda < 0$ and a simple minimum are required; $\lambda \geq 0$ is terminal.

Figure 12. Terminal boundary of the seed rule. Computed rule. For a nonempty complement, $\lambda < 0$ and a simple minimum are required; $\lambda \geq 0$ is terminal. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

The canonical dynamics is total only after adjoining explicit terminal outcomes.

$$U \sim_{\text{BFG}} : X_{\Xi} \rightarrow \backslash (X_{\Xi} / \text{Unitary}) \backslash \text{disjoint}_{\text{union}} T_{\text{terminal}} \backslash (53)$$

Code	Trigger	Meaning
T_sel0	$\text{rank}(\text{QP})=0$	No selected persistent carrier
T_w1	$m_{\text{sel}}=0$	First witness selection fails
T_load	$\Lambda_C \leq 0$ or $\Lambda_B \leq 0$	Positive dual-load class not satisfied
T_seednone	$E \neq 0$ and $\lambda_{\min} \geq 0$	Required negative seed absent
T_seeddeg	$E \neq 0$ and negative lowest eigenvalue is multiple	No unique rank-one seed

Table 2. Canonical classification and definitions.

The no-seed branch exists only when $E=0$. If $E \neq 0$, a simple strictly negative lowest eigenvalue is required; otherwise the event terminates.

$$E=0: \rho_{\text{seed}}=0, \Pi_{\text{seed}}=0 \quad (54a)$$

$$E \neq 0, \lambda_{\min} \geq 0: T_{\text{seednone}} \quad (54b)$$

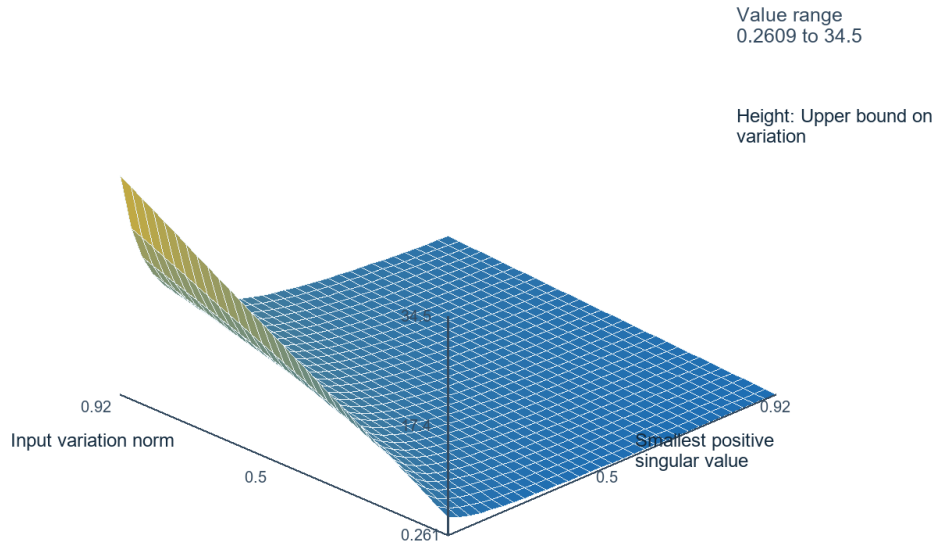
$$E \neq 0, \lambda_{\min} < 0 \text{ multiple}: T_{\text{seeddeg}} \quad (54c)$$

No arbitrary tie-break vector is introduced at a multiply degenerate negative minimum.

15 Constant-rank polar differential

Polar sensitivity bound

COMPUTED BOUND



3 times $\|\delta X\|_2$ divided by $\sigma_{\min,+}$; an upper bound is not an attained response.

Figure 13. Polar sensitivity bound. Computed bound. 3 times $\|\delta X\|_2$ divided by $\sigma_{\min,+}$; an upper bound is not an attained response. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Let $X=TH$ be a C^1 curve on a constant-rank stratum. Choose the canonical polar factor with $\ker(T)=\ker(X)$. Set $E_X=\delta X$, $H=(X^\dagger X)^{1/2}$, $S_- = T^\dagger T$ and $S_+ = TT^\dagger$. The following derivatives are tangent derivatives on that stratum:

$$(I - S_+) E_X (I - S_-) = 0 \quad (55)$$

The skew-Hermitian connection is supported inside the source: $\Omega = S_- \Omega S_-$. On that support it is uniquely fixed by the Sylvester equation:

$$H \Omega + \Omega H = S_- (T^\dagger E_X - E_X^\dagger T) S_- \quad (56)$$

The polar derivative is:

$$\delta T = T \Omega + (I - S_+) E_{XH}^\dagger + T H^\dagger T^\dagger E_X (I - S_-) \quad (57)$$

and the support derivatives are:

$$\delta S_+ = (I - S_+) E_{XH}^\dagger T^\dagger + T H^\dagger E_X^\dagger (I - S_+) \quad (58)$$

$$\delta S_- = H^\dagger T^\dagger E_X (I - S_-) + (I - S_-) E_X^\dagger T H^\dagger \quad (59)$$

A conservative bound is:

$$\|\delta T\|_2 \leq 3 \|E_X\|_2 / \sigma_{\min,+}(X) \quad (60)$$

This is consistent with the general sensitivity of polar factors to small singular values [4,5]. A diverging upper bound as $\sigma_k \rightarrow 0$ does not imply that every admissible curve diverges; it signals loss of a uniform local bound.

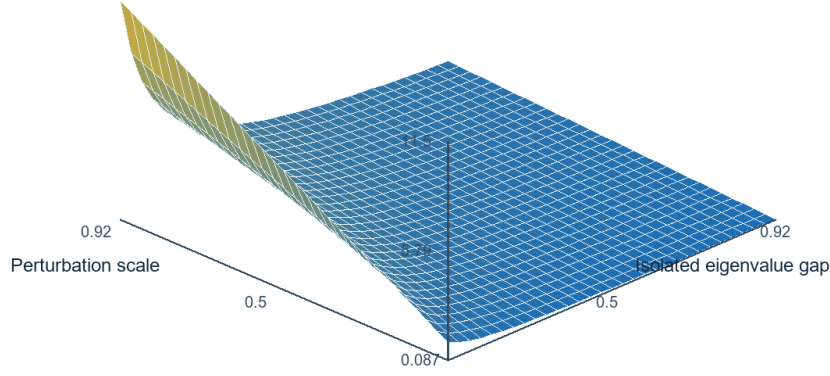
16 Seed differential

Seed eigenprojector sensitivity

CONDITIONAL PERTURBATION SCALE

Value range
0.08696 to 11.5

Height: Gap sensitivity
scale



Illustrates inverse-gap scaling of the reduced resolvent; not a full derivative of an arbitrary kernel.

Figure 14. Seed eigenprojector sensitivity. Conditional perturbation scale. Illustrates inverse-gap scaling of the reduced resolvent; not a full derivative of an arbitrary kernel. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

With $E = I - S_4^+$, choose a horizontal orthonormal frame Z_E :

$$\delta Z_E = (\delta E) Z_E \quad (61)$$

Then:

$$\delta H_E = Z_E^\dagger [(\delta E) K_+ + \delta K_+ + K_+ (\delta E)] Z_E \quad (62)$$

For a simple negative eigenvalue λ with $\pi = vv^\dagger$:

$$\delta \lambda = \text{tr}(\pi \delta H_E) \quad (63)$$

$$R_\lambda = (H_E - \lambda I)^+; R_\lambda v = 0 \quad (64)$$

$$\delta \pi = -R_\lambda (\delta H_E) \pi - \pi (\delta H_E) R_\lambda \quad (65)$$

$$\delta \Pi_{\text{seed}} = (\delta E) \Pi_{\text{seed}} + \Pi_{\text{seed}} (\delta E) + Z_E (\delta \pi) Z_E^\dagger \quad (66)$$

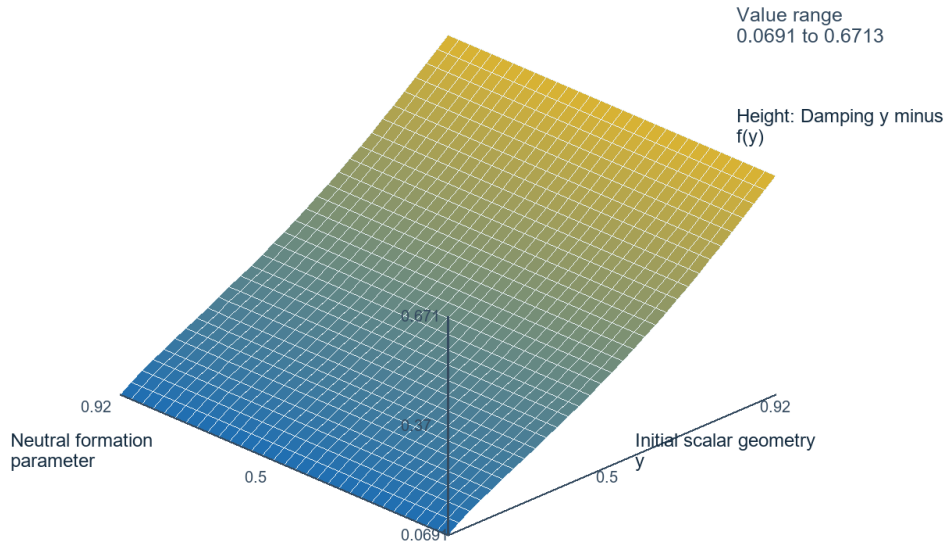
$$\delta \rho_{\text{seed}} = -\kappa_F [(\delta \lambda) \Pi_{\text{seed}} + \lambda \delta \Pi_{\text{seed}}] \quad (67)$$

At complement dimension one, $R_\lambda = 0$.

17 Endogenous geometric feedback and stability

Geometric self feedback and damping

EXACT SCALAR KERNEL



$P=W=1$ and $F>0$; $f(y)=2y^2/[(1+y)^2(1+y^2)]$; K and F remain neutral.

Figure 15. Geometric self feedback and damping. Exact scalar kernel. $P=W=1$ and $F>0$; $f(y)=2y^2/[(1+y)^2(1+y^2)]$; K and F remain neutral. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

The closed internal feedback loop is:

$$Y_k \rightarrow G_k / P_{G,k} \rightarrow \text{Loads} \rightarrow \alpha_k, \beta_k \rightarrow A_k \rightarrow \Sigma_k \rightarrow Y_{k+1} \quad (68)$$

This is a genuine endogenous geometric feedback: the current Gram geometry changes the analysis that produces the next Gram geometry. It is not, by itself, a theorem of full-state contraction.

Ambient bound. Let $D=Z^\dagger GZ$. Since $GC=I$ and $GB=Y$, $P_G C=Z D^{-1} Z^\dagger$ and $P_G B=Z D^{-1} Z^\dagger Y$. The first map has range exactly $\text{Ran}(P_{\text{sel}})$. Positive α gives $\text{Ran}(P_{\text{sel}}) \subset \text{Ran}(A^\dagger)$, proving the R_4 identities and $m_4=1$.

Let η be the largest selected Y eigenvalue; $\eta < 1$ and $D \geq I$. Then $\|P_G C\|_2 \leq 1$ and $\|P_G B\|_2 \leq \eta$. Hence $\|A\|_2^2 \leq \alpha + \beta \eta^2 < 1$. All reduced singular values are positive, so $0 < Y_+ < I$. This bounds the successor Gram operator, not the full endogenous Jacobian.

After a nonterminal event $Q_+=I$ and the successor witness is supported on P_+ . The following selection is $T_{\text{sel}} = P_+$ and has mass one. Thus every subsequent nonterminal event preserves original witness mass. The model cannot represent arbitrary repeated lossy disturbances without an additional declared perturbation or update rule.

On the independently checked scalar full-persistence branch:

$$f(y) = \frac{2y^2}{(1+y)^2(1+y^2)} \quad (69)$$

For $y > 0$:

$$y - f(y) = y[(1+y)^2(1+y^2) - 2y] / [(1+y)^2(1+y^2)] > 0 \quad (70)$$

Hence, on the nonterminal selection-first branch $0 < y < 1$:

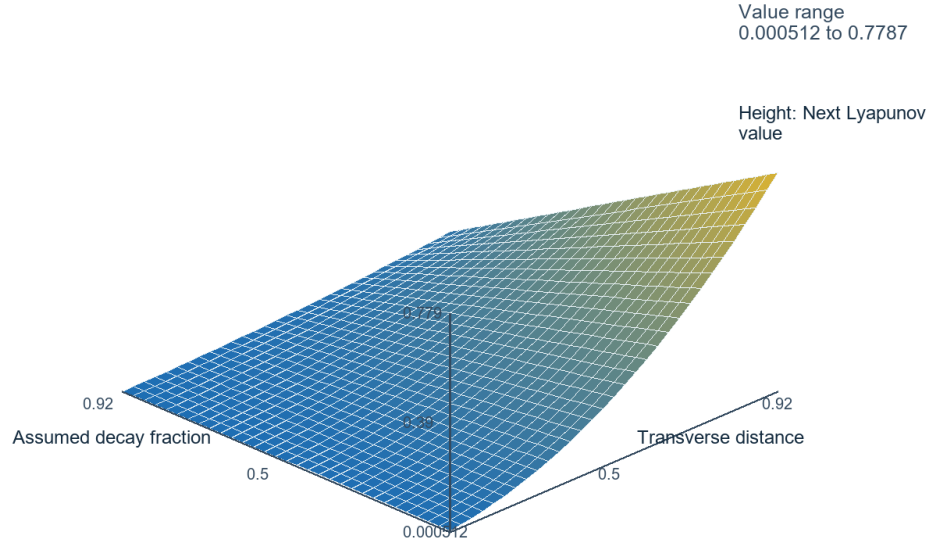
$$0 < f(y) < y \quad (71)$$

The same branch contains neutral K/F directions in which $K_+ = K$ and $F_+ = F$. Therefore no strict contraction theorem for the complete tuple can hold in every fixed positive product norm on that branch. Positivity preservation, scalar damping, transverse contraction, and full-state contraction are distinct statements.

18 Discrete and continuous Lyapunov formulations

Conditional transverse Lyapunov decay

CONDITIONAL STABILITY MODEL



Assumes $V = z^2$ and $V_{\text{next}} = (1-r)V$; no such global BFG certificate is constructed here.

Figure 16. Conditional transverse Lyapunov decay. Conditional stability model. Assumes $V = z^2$ and $V_{\text{next}} = (1-r)V$; no such global BFG certificate is constructed here. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Let M be an invariant neutral set and $\|z\|$ the quotient distance to M , or a specified transverse coordinate with equivalent distance bounds. On a forward-invariant, fixed-dimensional regular region R with all forward iterates defined, assume positive constants c_1, c_2, c_3 and:

$$c_1 \|z\|^2 \leq V(\Xi) \leq c_2 \|z\|^2 \quad (72)$$

$$\Delta V = V(U_{\text{BFG}}(\Xi)) - V(\Xi) \leq -c_3 \|z\|^2 \quad (73)$$

These are assumptions on a chosen V and region, not a constructed stability certificate for the BFG kernel. They imply exponential transverse decay in event count when the distance bounds hold uniformly: $V_{k+1} \leq (1-c_3/c_2) V_k$. The inequality requires a nonnegative compatible factor, with the zero-factor case giving immediate arrival. Existence, forward invariance and quotient regularity must be checked separately.

A continuous projected model is a different object:

$$d\Psi/dt = F_\Psi(\Psi) \quad (74)$$

for which:

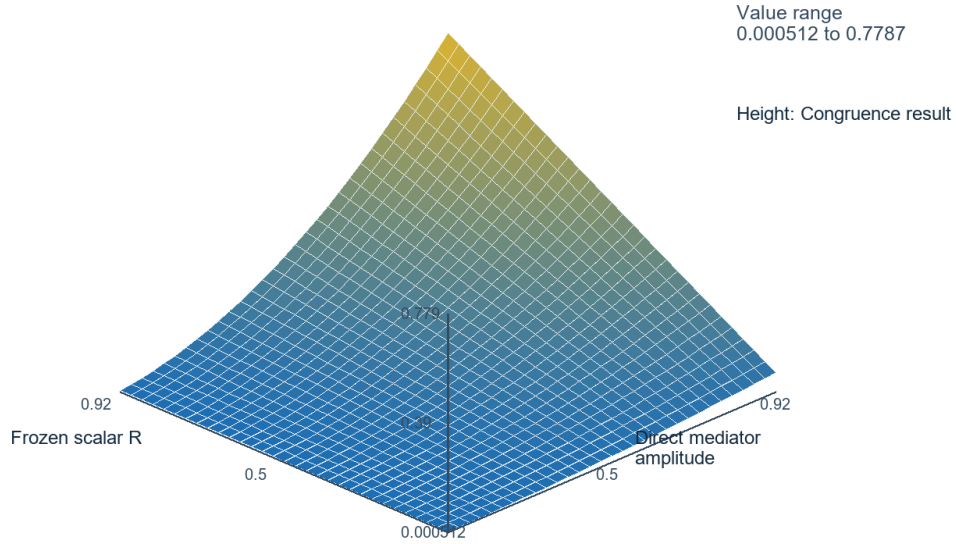
$$dV/dt = DV(\Psi) F_\Psi(\Psi) \quad (75)$$

A clock does not by itself construct F_ψ . Viability theory supplies the appropriate equation for state constraints and viable trajectories [9].

19 Mediated recursive closure

Mediator amplitude and power

COMPUTED MEDIATOR CONTROL



$N_B R N_B^\dagger = n^2 R$; amplitude mediation uses $N_A = n^2$, not $N_A = n$.

Figure 17. Mediator amplitude and power. Computed mediator control. $N_B R N_B^\dagger = n^2 R$; amplitude mediation uses $N_A = n^2$, not $N_A = n$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

For a positive semidefinite self-adjoint mediator N_A on the declared operator space, its unique positive square root gives the amplitude-mediated class:

$$R_C^A = N_A^{1/2} R N_A^{1/2} \quad (76)$$

The direct minimal class is:

$$R_C^B = N_B R N_B^\dagger \quad (77)$$

At $R=I$ the second class gives $N_B N_B^\dagger$, not N_B^2 unless N_B is self-adjoint. The two classes coincide under the explicit identification $N_B = N_A^{1/2}$. The conjugate-closure source line is associated with the Wende–Leighton precursor [2].

A spectral-radius condition is informative but not sufficient for existence by itself:

$$\rho(R_C) < 1, \rho(R_C) \approx 1, \rho(R_C) > 1 \quad (78)$$

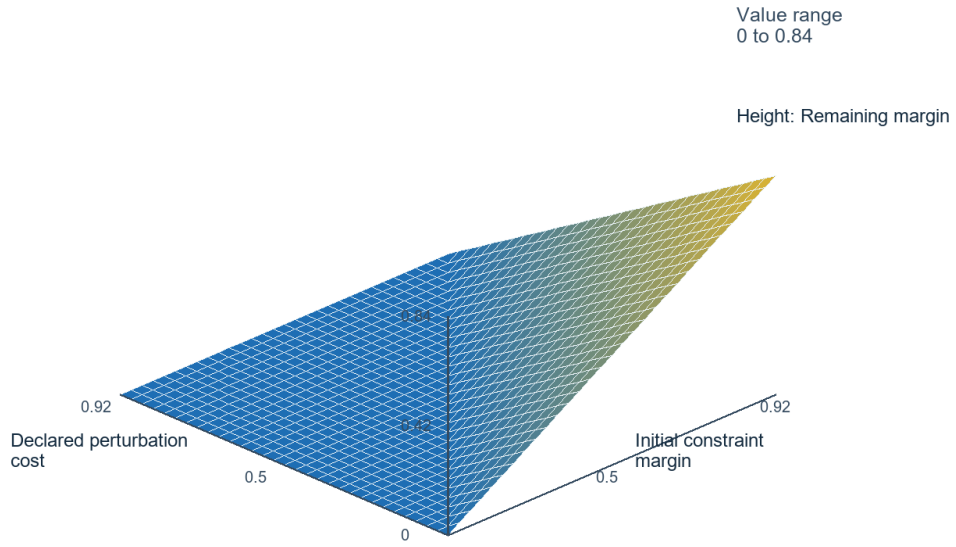
These regimes respectively suggest subcritical damping, near-neutral persistence, and possible amplification only after the operator space, norm, Jordan structure, and admissibility conditions are specified.

A fixed finite linear operator with spectral radius below one has powers tending to zero. This frozen-operator statement does not imply $\|D U_{BFG}\| < 1$ for an endogenously changing nonlinear kernel. Norm bounds and Jordan structure determine quantitative transient estimates.

20 Admissibility and viability

Finite horizon viability gate

SCHEMATIC CONSTRAINT MODEL



A margin diagram only; actual viability requires the unique nonterminal BFG trajectory and specified gates.

Figure 18. Finite horizon viability gate. Schematic constraint model. A margin diagram only; actual viability requires the unique nonterminal BFG trajectory and specified gates. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$S_{\text{adm}} = S_{\text{balance}} \cap S_{\text{feedback}} \cap S_{\text{spectral}} \cap S_{\text{interface}} \cap S_{\text{regular}} \cap S_{\text{stability}} \quad (79)$$

Static membership in S_{adm} is not identical to forward invariance. The viability kernel is:

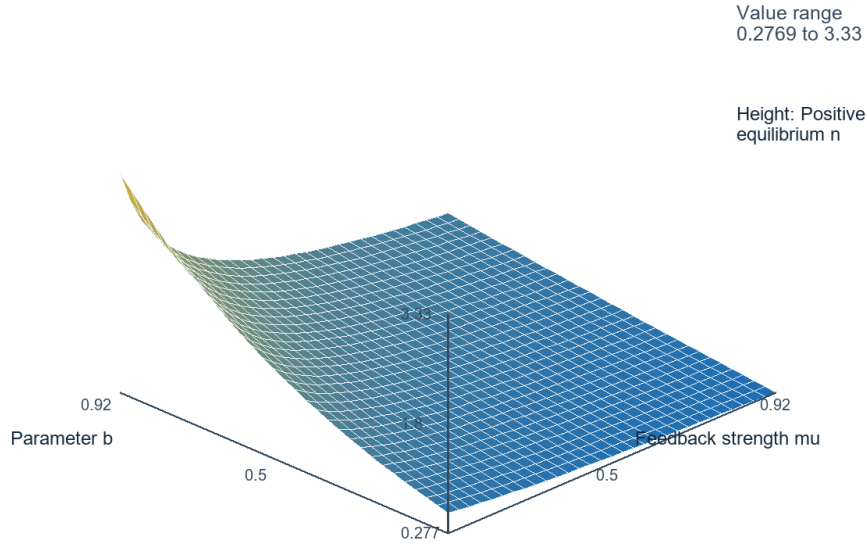
$$K_V = \{x_0 : U_{\text{BFG}}^k(x_0) \text{ lies in } S_{\text{adm}} \text{ and is nonterminal for every } k \geq 0\} \quad (80)$$

This is the viability set of the deterministic kernel. Terminal outcomes count as failure to persist. For a finite horizon restrict k to that horizon. An existential choice among trajectories requires an explicitly controlled or set-valued extension. The gates in (79) are an organizational list; each application must specify the actual constraint sets [9].

21 Scalar mediator normal form

Scalar normal form equilibrium

COMPUTED CONDITIONAL ODE



$a=1.5, \mu>0, b<1$; this positive branch exists because $a>1-b$.

Figure 19. Scalar normal form equilibrium. Computed conditional ode. $a=1.5, \mu>0, b<1$; this positive branch exists because $a>1-b$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$dq/dt = a - (1 - b + \mu n)q \quad (81)$$

$$\varepsilon dn/dt = q - 1 - n \quad (82)$$

At a positive fixed point $q^* = 1 + n^*$ and $r^* = 1 - b + \mu n^* = a/q^* > 0$. The Jacobian has:

$$\text{tr } J^* = -r^* - 1/\varepsilon < 0 \quad (83)$$

$$\det J^* = \frac{r^* + \mu q^*}{\varepsilon} > 0 \quad (84)$$

Conditional on existence of a positive fixed point, $a, \mu, \varepsilon > 0$ give negative trace and positive determinant and hence local asymptotic stability. For $n^* > 0$, existence requires $a > 1 - b$: the fixed-point equation $a = (1 + n^*)(1 - b + \mu n^*)$ has a unique positive solution with $r^* > 0$ under this condition. In particular $b < 1$ and small a need not admit a positive fixed point. Varying ε does not create a Hopf threshold on this unchanged positive branch.

For the additional quadratic overconstraint model $d(n) = 1 - \zeta n^2$ in the overcritical regime:

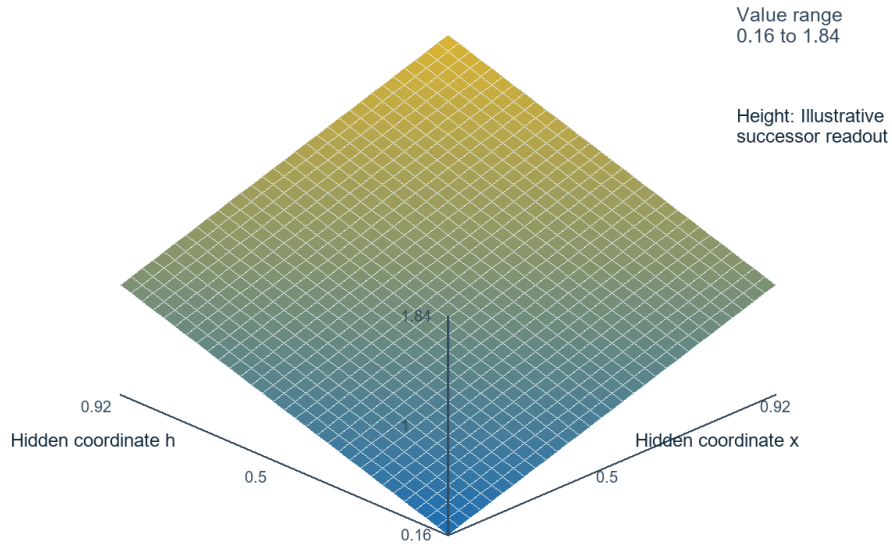
$$n_{\min} = (b - 1)/\mu; n_{\max} = \sqrt{(1 - d_{\min})/\zeta} \quad (85)$$

with $b > 1, \mu > 0, \zeta > 0$ and $d_{\min} < 1$. The open static corridor exists when $n_{\min} < n_{\max}$. This static corridor is not the viability kernel.

22 Functional state Psi and factor dynamics

A projection must close its fibers

SCHEMATIC COUNTERMODEL



Projection $\Pi(x,h)=x$ with successor $x+h$ is not closed; this is a fiber test, not a BFG bridge.

Figure 20. A projection must close its fibers. Schematic countermodel. Projection $\Pi(x,h)=x$ with successor $x+h$ is not closed; this is a fiber test, not a BFG bridge. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$\Psi = (\varphi_{\text{fun}}, I_{\text{fun}}, A_{\text{fun}}, G_{\text{fun}}, \Omega_{\text{fun}}, T_{\text{fun}}, S_{\text{fun}}, M_{\text{fun}}, C_{\text{fun}}) \quad (86)$$

A closed discrete readout dynamics requires a typed projection $\Pi: X_{\Xi} \rightarrow X_{\Psi}$ satisfying:

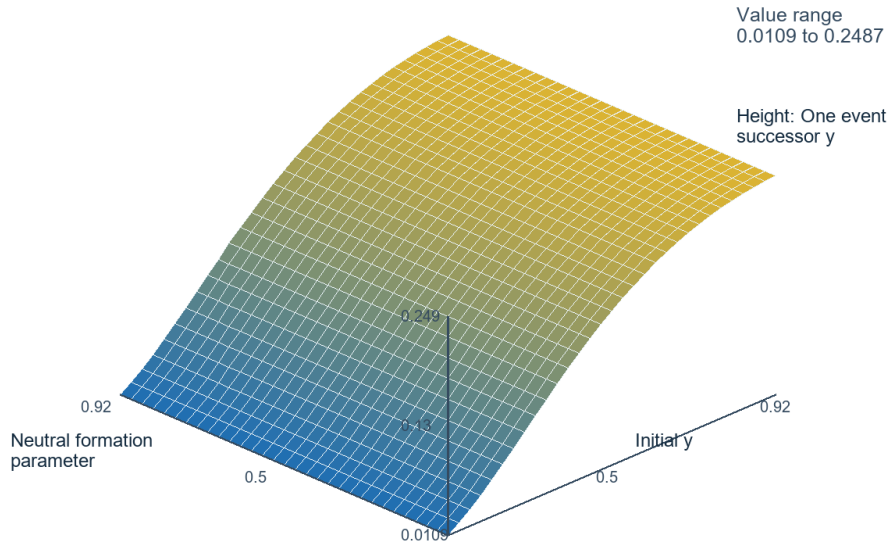
$$\Pi(x) = \Pi(x') \Rightarrow \Pi(U_{\text{BFG}} x) = \Pi(U_{\text{BFG}} x') \quad (87)$$

The subscripts distinguish functional roles from kernel operators and from Φ_0 . φ_{fun} may equal a selected scalar order readout only under a specified bridge. The fiber criterion guarantees a well-defined discrete factor on $\Pi(X_{\Xi})$, not smoothness or physical identification. Terminality must also be constant on fibers. No such concrete Π is constructed here.

23 Event time, model time, and physical time

Event count without calibrated seconds

EXACT SCALAR KERNEL



One reclosure event is counted; no conversion to physical seconds is claimed.

Figure 21. Event count without calibrated seconds. Exact scalar kernel. One reclosure event is counted; no conversion to physical seconds is claimed. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

The kernel index k is an event/reclosure count:

$$k \in \mathbb{N}_0 \setminus \{88\}$$

A physical clock is an additional positive map:

$$T_{\text{clock}} : X_{\Xi} / \text{Unitary} \rightarrow \mathbb{R}_{>0} \quad (89)$$

$$t_k = t_0 + \sum_{j < k} T_{\text{clock}}([\Xi_j]) \quad (90)$$

For a given autonomous continuous flow ϕ_F^t , the correct compatibility condition is:

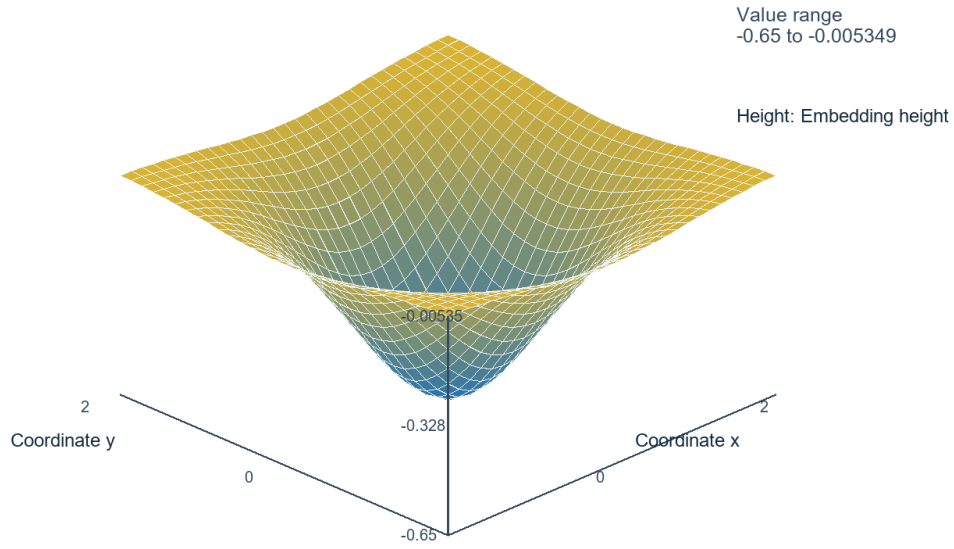
$$\Pi(U_{\text{BFG}} \Xi) = \phi_F^{T_{\text{clock}}([\Xi])}(\Pi(\Xi)) \quad (91)$$

A clock is a quotient-invariant, independently calibrated extra map. Positive durations alone do not exclude Zeno accumulation: $\sum_k T_k = 2^k$ has finite sum. Unbounded physical continuation additionally requires $\sum_k T_k = \infty$. A continuous generator and the endpoint-flow equality require independent construction. Clock constancy on Π -fibers is necessary only when a projected clock depending on Ψ alone is demanded.

24 From internal Gram geometry to spacetime

A curved geometry bridge illustration

SCHEMATIC EMBEDDING



An illustrative surface is not a Lorentz metric, gravitational field equation or prediction.

Figure 22. A curved geometry bridge illustration. Schematic embedding. An illustrative surface is not a Lorentz metric, gravitational field equation or prediction. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

The internal metric $G=I+Y$ is positive definite. A Lorentzian spacetime metric belongs to a different type of object: it requires a manifold, dimension, signature, connection, and curvature. One possible additional completion is the rank-one disformal form studied in gravitational theory [13]:

$$g_{\mu\nu} = \eta_{\mu\nu} + \lambda \partial_{\mu} \phi \partial_{\nu} \phi \quad (92)$$

with η signature $(-, +, +, +)$. Define $X_{\eta} = \eta^{\mu\nu} \partial_{\mu} \phi \partial_{\nu} \phi$. Then:

$$\det g = \det \eta \cdot (1 + \lambda X_{\eta}) \quad (93)$$

$$g^{\mu\nu} = \eta^{\mu\nu} - \lambda \partial^{\mu} \phi \partial^{\nu} \phi / (1 + \lambda X_{\eta}) \quad (94)$$

The Lorentz-connected nondegeneracy condition is:

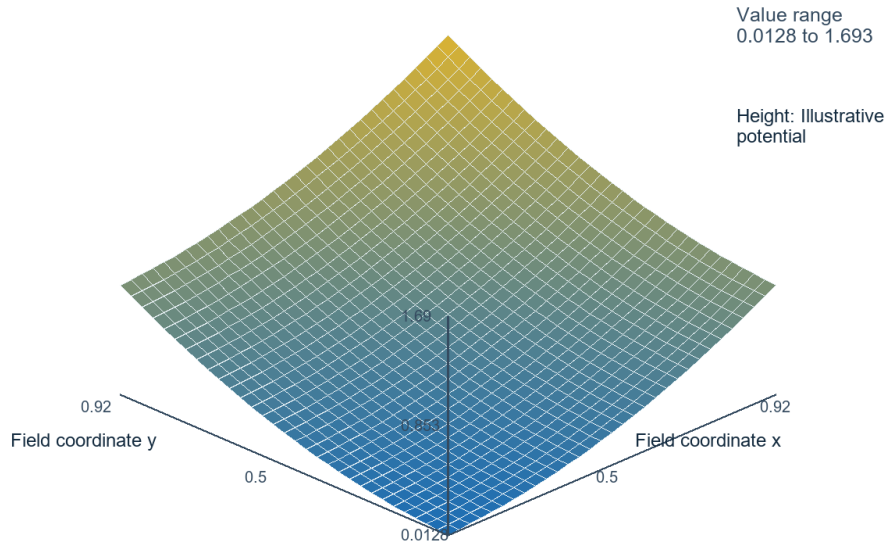
$$1 + \lambda X_{\eta} > 0 \quad (95)$$

For $\eta = \text{diag}(-1, 1, 1, 1)$ in Cartesian coordinates, $\det \eta = -1$. In natural units with $[\phi] = M$ and $[\partial \phi] = M^2$, $[\lambda] = M^{-4}$. This rank-one completion is additional physics; it does not derive spacetime from the positive Gram operator.

25 Gravitational completion

Geometric feedback requires dynamics

ASSUMED POTENTIAL MODEL



$V=x^2+y^2$ is a pedagogical potential; stability of spacetime must follow from a specified action and bridge.

Figure 23. Geometric feedback requires dynamics. Assumed potential model. $V=x^2+y^2$ is a pedagogical potential; stability of spacetime must follow from a specified action and bridge. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

A canonical real scalar field with $(-+++)$ convention can be coupled through:

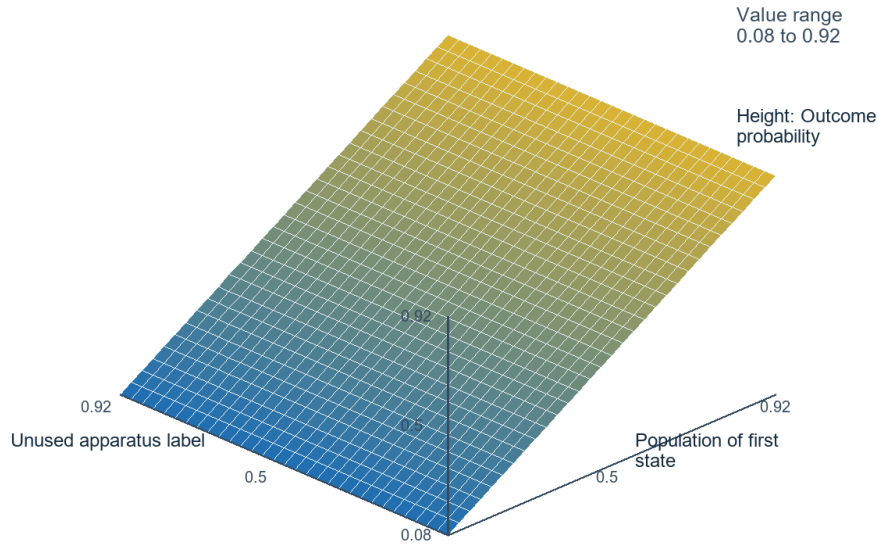
$$S = \int \sqrt{-g} \left[\frac{M_{\text{pl}}^2}{2} R - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - V(\phi) \right] d^4x + S_m \quad (96)$$

This is an assumed Einstein–scalar action. Its independent-field variation requires compactly supported variations or appropriate boundary data and gravitational boundary terms. Matter variation must be included when S_m depends on ϕ . Constraining $g=g[\phi]$ before variation gives a different chain-rule theory. The finite kernel supplies neither this action nor Newton’s constant, a Lorentz manifold or a physical scalar field.

26 Quantum bridge

Quantum state identification remains a bridge

CONDITIONAL QUANTUM MODEL



For a calibrated $E_0=|0\rangle\langle 0|$, $p=\text{tr}(E_0 \rho)$; the state-identification map is additional physics.

Figure 24. Quantum state identification remains a bridge. Conditional quantum model. For a calibrated $E_0=|0\rangle\langle 0|$, $p=\text{tr}(E_0 \rho)$; the state-identification map is additional physics. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

A complete quantum bridge would be:

$$\Pi_Q : \Xi \rightarrow (H_Q, \rho_Q, A_Q, H_{\text{hat}}) \quad (97)$$

where H_Q is a Hilbert space, ρ_Q a density operator, A_Q an observable algebra, and H_{hat} a Hamiltonian.

Quantum operations are naturally represented by completely positive maps and instruments [11,12]. Therefore nonunitarity of the finite BFG kernel does not by itself exclude a quantum interpretation; actual instrument compatibility must be demonstrated.

$$H_A : \rho_Q^A = \rho_W \quad (98)$$

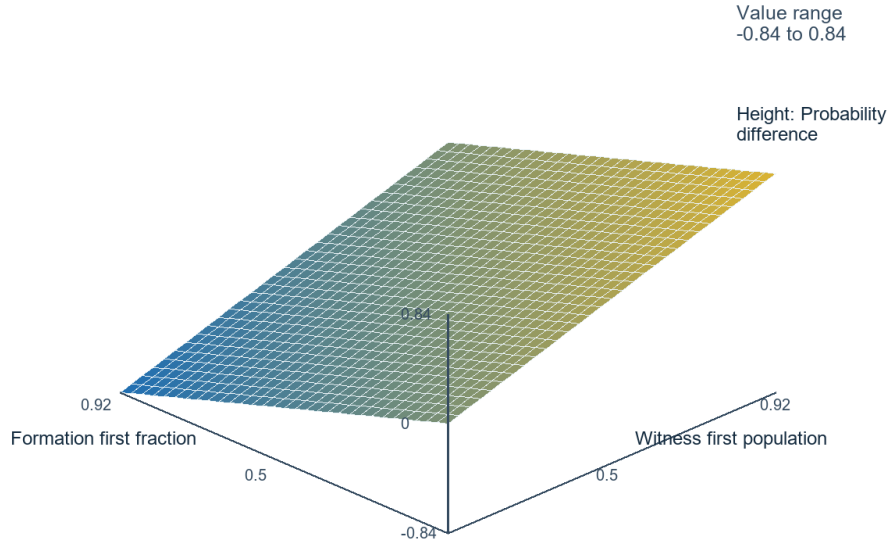
$$H_B : \rho_Q^B = \rho_F / \text{tr}(\rho_F) \quad (99)$$

Neither is a theorem of the finite kernel.

27 Finite quantum discrimination example

Competing quantum bridge predictions

COMPUTED BRIDGE COMPARISON



At (0.75,0.25), hypotheses A and B differ by 0.5; no actual measured outcomes are used.

Figure 25. Competing quantum bridge predictions. Computed bridge comparison. At (0.75,0.25), hypotheses A and B differ by 0.5; no actual measured outcomes are used. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

$$K = -I_2; Y = \text{diag}(1/2, 1/4); P = I_2 \setminus (100)$$

$$\rho_F = \text{diag}(1, 3); \rho_W = \text{diag}(3/4, 1/4) \setminus (101)$$

Then $Q = T_{\text{sel}} = P_{\text{sel}} = I_2$ and $m_{\text{sel}} = 1$.

$$C = \text{diag}(2/3, 4/5); B = \text{diag}(1/3, 1/5); G = \text{diag}(3/2, 5/4) \setminus (102)$$

$$\Lambda_C = 46/15; \Lambda_B = 19/60 \setminus (103)$$

Both loads are positive, so the analysis operator has full column rank. V is unitary, $X_{R4} = V^\dagger$, $S_4^{\wedge-} = S_4^{\wedge+} = I_2$, $m_4 = 1$, and $E = 0$.

For an externally fixed effect $E_0 = |0\rangle\langle 0|$:

$$p_A = \text{tr}(E_0 \rho_W) = 3/4 \setminus (104)$$

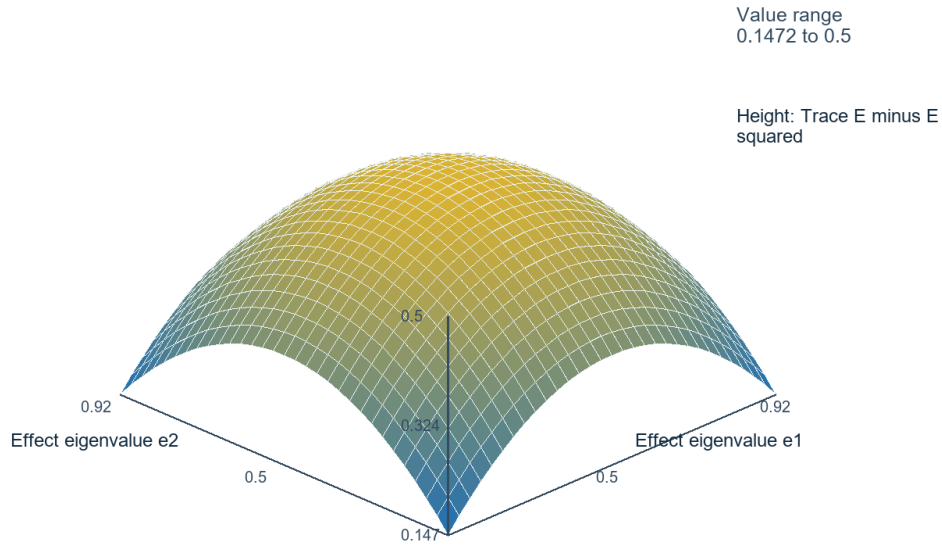
$$p_B = \text{tr}(E_0 \rho_F) / \text{tr}(\rho_F) = 1/4 \setminus (105)$$

The two bridge hypotheses are distinguishable in principle. E_0 is not the actual R_4 success effect of this example: here $S_4^{\wedge-} = I$ and the R_4 success mass is one.

28 R4 and quantum instruments

A general effect need not be projective

COMPUTED EFFECT TEST



A direct R4 effect is a projector. Generic positive effects with interior eigenvalues fail that test.

Figure 26. A general effect need not be projective. Computed effect test. A direct R4 effect is a projector. Generic positive effects with interior eigenvalues fail that test. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

For a fixed independently specified T_4 , $\rho \rightarrow T_4 \rho T_4^\dagger$ is completely positive, with effect $T_4^\dagger T_4 = S_4^\wedge$. Direct identification with a measured Kraus success operator therefore requires a projective effect and compatible transport. A general POVM effect need not be a projector [11].

Because T_4 is endogenously state-dependent,

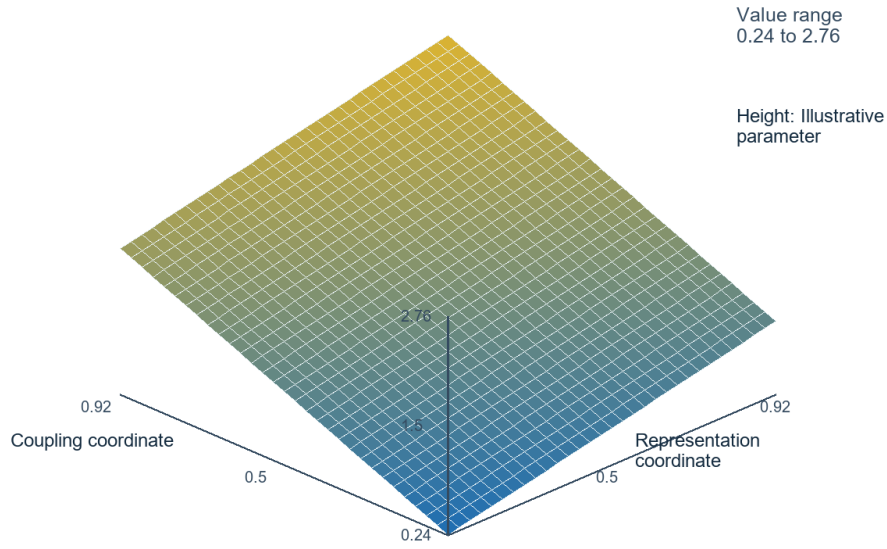
$$T_4 = T_4 \backslash (\Xi) \quad \backslash (106)$$

the family $\rho \rightarrow T_4(\Xi) \rho T_4(\Xi)^\dagger$ is not automatically one linear CP channel on arbitrary inputs. It can be a classically conditioned family only if Ξ is independently identified.

29 Gauge and particle sector

Particle hierarchy requires parameters

SCHEMATIC DEPENDENCY



The surface labels a parameter dependency only; no Standard Model mass or coupling is derived.

Figure 27. Particle hierarchy requires parameters. Schematic dependency. The surface labels a parameter dependency only; no Standard Model mass or coupling is derived. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

The Standard Model is experimentally organized around $SU(3)_C \times SU(2)_L \times U(1)_Y$ and measured particle properties summarized by the Particle Data Group [14]. A BFG particle bridge must map the finite recursive state to field representations, charges, couplings, symmetry breaking, masses, and mixing data.

$$\Pi_{SM} : \Xi \rightarrow \{\text{gauge fields, representations, couplings, masses, mixings}\} \quad (107)$$

The current finite core does not determine these numerical quantities.

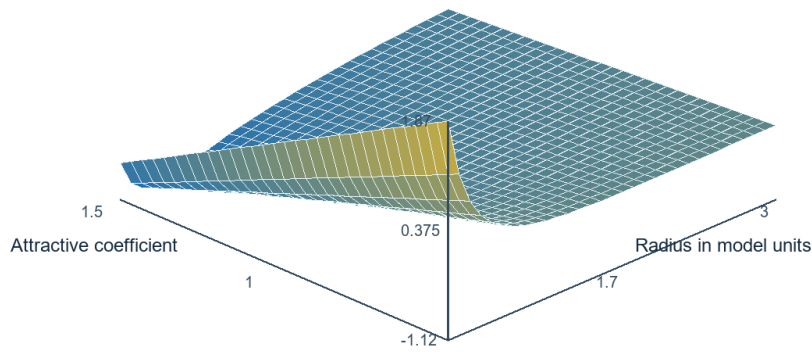
30 Atomic closure

Atomic localization and attraction

CONVENTIONAL CONDITIONAL MODEL

Value range
-1.125 to 1.875

Height: Variational energy



$E(r)=1/(2r^2)-a/r$; ordinary quantum scaling, not a new BFG atomic spectrum.

Figure 28. Atomic localization and attraction. Conventional conditional model. $E(r)=1/(2r^2)-a/r$; ordinary quantum scaling, not a new BFG atomic spectrum. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

A structural atom-level closure can be denoted:

$$\text{Atom} = \text{Cl}_N \backslash (\text{particles}, \text{interactions}) \quad (108)$$

but Cl_N is not a complete transition law until its state space and dynamics are given. Quantitative atomic spectra currently require ordinary quantum mechanics and electrodynamics. A familiar scale argument,

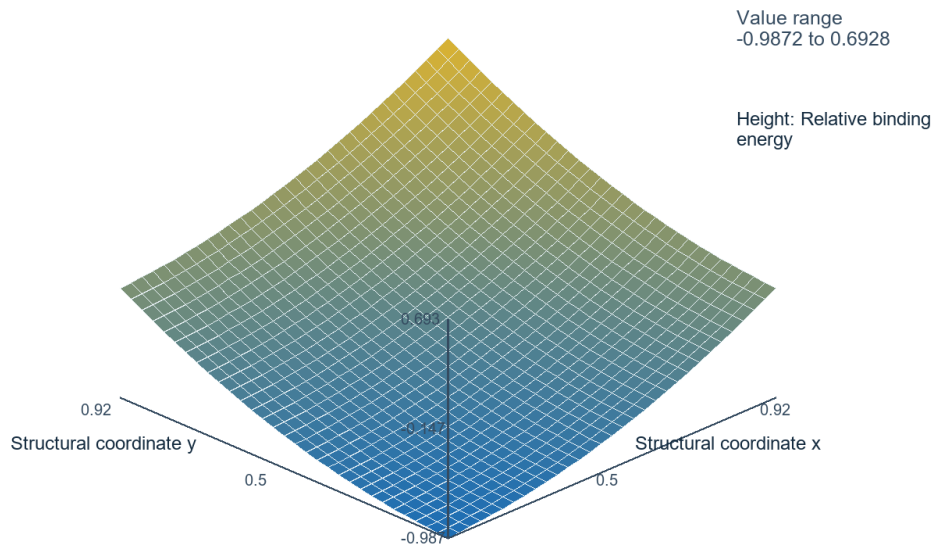
$$E(r) \sim \hbar^2 / (2\mu r^2) - e^2 / (4\pi \epsilon_0 r) \quad (109)$$

shows why the positive r^{-2} localization term can dominate the attractive r^{-1} term at short distance. This is a quantum variational scaling argument, not a derivation of $\hbar$ or e from the finite kernel.

31 Molecular and biological closure

Molecular binding criterion

SCHEMATIC BINDING LANDSCAPE

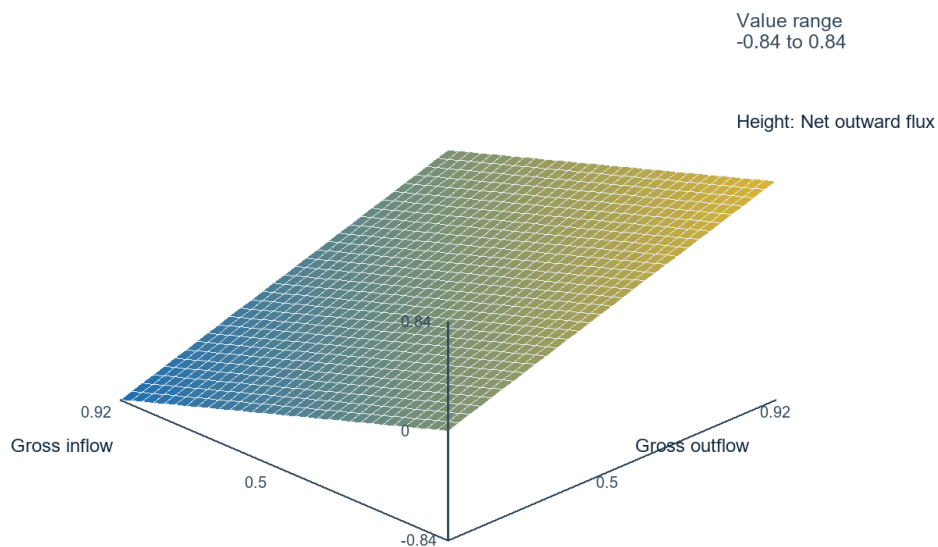


The illustrative energy x^2+y^2-1 marks a negative binding region; it is not a BFG molecular potential.

Figure 29. Molecular binding criterion. Schematic binding landscape. The illustrative energy x^2+y^2-1 marks a negative binding region; it is not a BFG molecular potential. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Biological boundary and gross flux

COMPUTED FLUX IDENTITY



$J_{\text{net}} = J_{\text{out}} - J_{\text{in}}$. Stationary net flux permits nonzero gross flows; no cell viability is inferred.

Figure 30. Biological boundary and gross flux. Computed flux identity. $J_{\text{net}} = J_{\text{out}} - J_{\text{in}}$. Stationary net flux permits nonzero gross flows; no cell viability is inferred. Horizontal and vertical coordinate ranges are the sampled grids; the

plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

A molecular stage may be schematically written:

$$M = Cl_{NM} \setminus (A_1, A_2) \setminus (110)$$

with a binding criterion $E(M) < E(A_1) + E(A_2)$, but a genuine bridge requires a typed molecular state and update. Turing's reaction-diffusion work is a canonical example of local dynamics generating higher spatial organization [15], while autopoiesis supplies an influential organizational language for self-maintaining living systems [16].

$$\text{molecule} \rightarrow \text{macromolecule} \rightarrow \text{cell} \rightarrow \text{tissue} \rightarrow \text{organ} \rightarrow \text{organism} \setminus (111)$$

The ladder is not a proof merely because the labels are ordered. Each arrow requires a state space, transition rule, preservation/loss quantities, and causal/viability criterion.

For an oriented biological boundary:

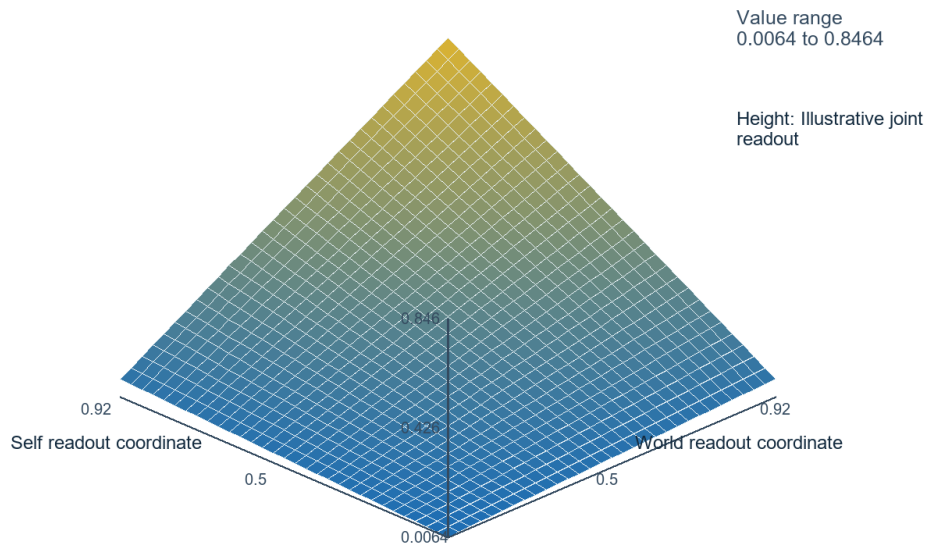
$$J_{\text{net}} = J_{\text{out}} - J_{\text{in}} \setminus (112)$$

Stationary net flux $J_{\text{net}} = 0$ means gross inflow equals gross outflow, not that both vanish.

32 Witness, world, self, and subject structure

World and self require operational maps

SCHEMATIC FUNCTIONAL MODEL



A product is only a visual aid. Π_W , Π_S , lineage and causal closure must be independently defined.

Figure 31. World and self require operational maps. Schematic functional model. A product is only a visual aid. Π_W , Π_S , lineage and causal closure must be independently defined. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Four notions are distinct: original witness mass w , carrier retention, formation retention and directed lineage. ρ_w is a kernel component; local m_{sel} and m_4 are computed event outputs and w and the transport history are separate trajectory records. They cannot generally be reconstructed from a normalized successor alone.

$$m_{\text{tot}} = m_{\text{sel}} m_4 \setminus (113)$$

Conditional normalization alone never proves full original-mass survival. Here $m_4=1$ is a theorem of the ambient analysis, whereas m_{sel} can be less than one at the initial event.

World and self require additional readouts:

$$W = \Pi_W(\Xi_{\text{history}}); S = \Pi_S(\Xi_{\text{history}}) \quad (114)$$

and a closed update:

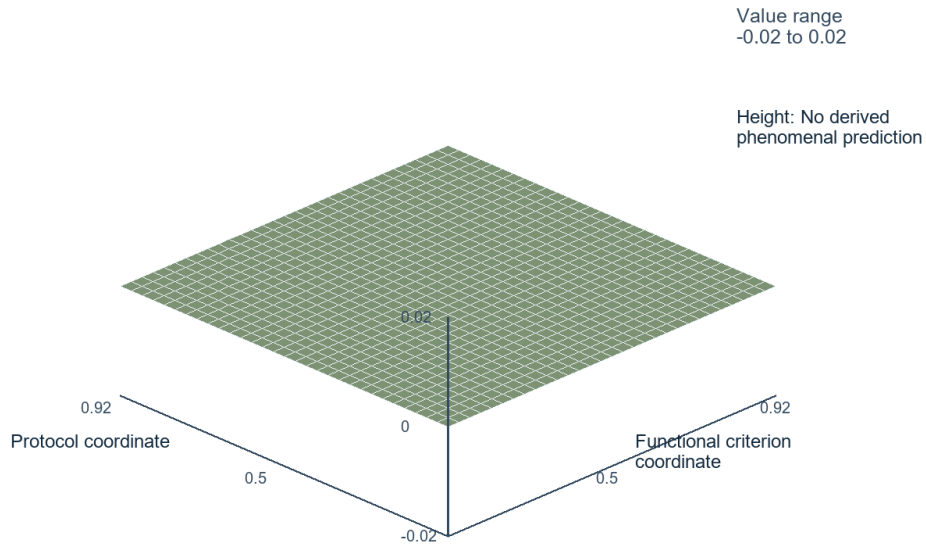
$$(W_{k+1}, S_{k+1}) = U_{WS}(W_k, S_k, u_k) \quad (115)$$

A subject criterion Γ_{subj} can only become operational after these maps, their causal roles, and measurement rules are fixed. The Wende–Pereira precursor motivates the distinction between functional/computational description and the stronger question of subjectivity [3]. Predictive/free-energy formulations provide a comparison point for world-model and self-regulation ideas [17], but do not validate the BFG bridge.

33 Phenomenal bridge

Phenomenality is not a matrix consequence

SCHEMATIC STATUS PANEL



The zero sheet denotes absence of a derived prediction, not absence of subjective experience.

Figure 32. Phenomenality is not a matrix consequence. Schematic status panel. The zero sheet denotes absence of a derived prediction, not absence of subjective experience. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Even a fully operational Γ_{subj} would not make phenomenality a mathematical theorem. The final identification would be an empirical-ontological hypothesis:

$$H_{\text{phen}} : \Gamma_{\text{subj}} = 1 \Leftrightarrow \text{phenomenal experience} \quad (116)$$

This hypothesis is separate from the finite-kernel mathematics and from the Level-0 ontological interpretation.

34 Canonical world-formula statement

The finite BFG world formula is the orbit of the quotient dynamics:

$$W_{\text{BFG}} = \{ \text{Orb}_{\text{Ubar}} \setminus ([\Xi_0]) \mid \text{each step is admissible or terminates by an explicit terminal code} \} \quad (117)$$

The nonterminal successor is:

$$[\Xi_k] \rightarrow \text{Selection} \rightarrow \text{metric loads} \rightarrow \text{ambient reduced SVD} \rightarrow [K_+, Y_+] \quad (118a)$$

$$[K_+, Y_+] \rightarrow \text{isometric R4} \rightarrow \text{witness/formation inheritance} \rightarrow \text{seed or terminal} \rightarrow [\Xi_{k+1}] \quad (118b)$$

This formulation is finite, stratified, gauge-covariant, and dimension-aware. It is a world formula in the mathematical sense of a state-plus-law architecture. The stronger statement that the physical universe is isomorphic to W_{BFG} is a physical hypothesis supplied by bridge laws and measurements.

35 Falsifiability and bridge discipline

A physical BFG bridge must be fixed before the holdout observations it is meant to predict. For a bridge B and preparation data D_{prep} :

$$\Xi = M_{\text{prep}} \setminus (D_{\text{prep}}) \quad (119)$$

$$P_{\text{BFG}} \setminus (D_{\text{future}} \mid D_{\text{prep}}, B) \quad (120)$$

must be computed without refitting B to holdout outcomes. The quantum A/B example illustrates this requirement. A negative result first falsifies the tested bridge hypothesis, not every mathematical statement of the finite kernel.

The same discipline applies to gravity, particle physics, biology, and subject structure. A bridge freely replaceable after every observation has no predictive content.

Identifiability is a separate requirement. Full column rank of a sensitivity Jacobian can provide a sufficient regular local criterion under suitable assumptions, but is not necessary for structural local uniqueness. The map $Y(\theta) = \theta^3$ is injective even though its derivative is zero at $\theta=0$. State the relevant equivalence symmetries and test local uniqueness of the actual preparation or bridge map; do not infer it solely from a first-order rank test.

36 Variable and operator register

Symbol	Type	Role	Canonical status
Φ_0	Level-0 object	pre-operational domain	bridge to Ξ not fixed
Ξ	five-component operator tuple	finite complete state	defined
K	Hermitian operator	kernel/formation geometry	defined
ρ_F	positive operator	formation	defined, not normalized
ρ_W	positive trace-one operator	witness	defined
Y	positive semidefinite operator	Gram/load geometry	defined
P	orthogonal projector	persistent carrier	defined
Q	spectral projector	selection	defined
T_{sel}	partial isometry	first transport	defined
P_G	G-orthogonal projector	metric selection	defined
Λ_C, Λ_B	nonnegative scalars	endogenous loads	defined
α, β	weights	analysis balance	dual-load branch
A	linear operator	analysis map	defined
U, Σ, V	reduced SVD factors	successor frame/spectrum	defined modulo gauge
T_4	partial isometry	R4 transport	defined

Symbol	Type	Role	Canonical status
$m_{\text{sel}}, m_4, m_{\text{tot}}$	$[0,1]$ scalars	witness survival	defined
E	projector	free target complement	defined
H_E	Hermitian compression	seed spectral problem	defined
P_{seed}	projector	seed support	simple negative branch
Ψ	nine-role functional state	readout architecture	$(\varphi_{\text{fun}}, I_{\text{fun}}, A_{\text{fun}}, G_{\text{fun}}, \Omega_{\text{fun}}, T_{\text{fun}}, S_{\text{fun}}, M_{\text{fun}}, C_{\text{fun}})$
T_{clock}	$X_{\Xi}/\text{Unitary} \rightarrow R_{\{>0\}}$	physical step duration	physical bridge
$g_{\mu\nu}$	Lorentz metric	spacetime geometry	physical completion
ρ_Q	density operator	quantum state	competing bridge hypotheses
Γ_{subj}	future operational gate	subject structure	not yet fixed
w	trajectory scalar	original witness mass	separate record; $w_{\text{next}} = w * m_{\text{sel}}$
q_{info}	scalar stock	φ times I	not selection Q
a_{info}	scalar injection rate	stock source	not analysis A

Table 3. Canonical classification and definitions.

37 Scope of proved statements

The following are mathematical consequences of the declared canonical class:

1. $G=I+Y$ is positive definite whenever $Y \geq 0$.
2. The positive dual-load ambient kernel gives $0 < Y_+ < I$ and $n_+ \leq n$.
3. The reduced-SVD convention fixes the active successor dimension $r = \text{rank}(A)$.
4. The unitary quotient removes common SVD phase and exactly degenerate block-basis ambiguity from the successor state.
5. m_{sel} is a survival mass in $[0,1]$; ambient R_4 has $m_4=1$ and $m_{\text{tot}} = m_{\text{sel}}$.
6. Hermiticity of K makes the seed compression Hermitian.
7. On the stated scalar full-persistence branch, $0 < f(y) < y$ for $0 < y < 1$.
8. The same branch prevents a universal strict full-state contraction theorem in fixed positive product norms because neutral K/F directions exist.
9. The positive fixed point of the scalar two-variable ODE is locally asymptotically stable for positive a, μ, ε under the stated branch conditions.

The following are not consequences of the finite kernel alone: a unique physical information measure, SI time, Lorentzian spacetime, Einstein dynamics, the Born rule, Standard-Model parameters, quantitative atomic/molecular chemistry, organismal closure, or phenomenal experience.

38 Discussion

The canonicalization changes the status of the finite kernel in an important way. Its main ambiguities are no longer left as informal choices. The successor space is the active reduced-SVD space; SVD basis freedom is quotient gauge; witness masses have a normalized probabilistic kernel interpretation; K is Hermitian; the seed has compatible units; and failure modes are named terminal outcomes. This makes the finite dynamics a genuine mathematical model rather than a collection of mutually suggestive formulas.

At the same time, the architecture remains layered. Internal Gram geometry is not silently promoted to spacetime. A kernel density operator is not silently promoted to a quantum state. A closure label is not silently promoted to a biological law. A world/self functional architecture is not silently promoted to phenomenality. Each promotion is an explicit bridge law that can be examined independently.

39 Conclusion

The Balance–Field Equation can be stated as a canonical stratified recursive dynamics:

$$[\Xi_{k+1}] = \text{Ubar}_{\text{BFG}}([\Xi_k]) \quad (121)$$

The stated model has an ambient analysis, canonical polar factors, positive dual loads and a seed rule that terminates when a nonempty complement lacks a simple negative minimum. It is defined on unitary equivalence classes. General consequences include $0 < Y_- < I$, $n_- \leq n$ and $m_4 = 1$; subsequent nonterminal events preserve witness mass. Scalar damping is a narrower geometric statement. Full regional contraction remains open and cannot hold on a region containing the neutral K/F control directions in a positive product norm. Physical universality requires additional fixed bridges and independent tests.

40 The BFG as a Fundamental Recursive World Equation

The canonical BFG world equation is not a shorthand for several unrelated laws. It is the single recursive evolution law on the quotient state space; spectral selection, metric projection, endogenous loading, reduced SVD, R4 transport, witness conditioning, formation inheritance, and seed formation are the explicit factorization of its evolution operator.

$$[\Xi_{k+1}] = \text{Ubar}_{\text{BFG}}([\Xi_k])$$

$$\text{Ubar}_{\text{BFG}} = U_{\text{seed}} \circ U_{\text{R4}} \circ U_{\text{SVD}} \circ U_{\text{load}} \circ U_{\text{metric}} \circ U_{\text{selection}}$$

Its composite operator structure is therefore a strength rather than an incompleteness: the state carries the coupled structures required for recursive persistence, while the factorization exposes every internal operation that produces the successor. The remaining question is no longer whether the BFG has a world equation; it is whether the physical measurement maps below identify this mathematical orbit with the observed world.

41 Physical Observables and Measurement Map

A physical world equation must connect internal state variables to independently measured quantities. The canonical measurement layer is therefore a family M_{phys} of typed readout maps. SI units follow the International System of Units; particle observables are to be compared with independently tabulated experimental quantities rather than fitted labels.

$$O_{\text{phys}} = M_{\text{phys}}([\Xi], \text{calibration, apparatus})$$

A BFG physical bridge is accepted only after the readout, calibration constants, preparation protocol, and uncertainty model are fixed independently of the holdout observations.

Symbol or observable	Complete specification
Physical time Δt	SI / standard unit: s Experimental measurement: atomic/optical clock interval BFG readout target: $T_{\text{clock}}(\Xi)$ Required bridge: $M_t: \Xi \rightarrow \mathbb{R}_+$ Calibration / nuisance: clock scale and synchronization Falsification criterion: predicted intervals inconsistent with calibrated clock data
Length / displacement	SI / standard unit: m Experimental measurement: interferometry, ranging BFG readout target: metric distance from $g_{\mu\nu}$ Required bridge: M_g plus coordinate/rod prescription Calibration / nuisance: coordinate gauge, apparatus response Falsification criterion: predicted invariant distances disagree with measurements

Symbol or observable	Complete specification
Proper time	SI / standard unit: s Experimental measurement: clock comparison along worldlines BFG readout target: $c^{-1} \int \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$ Required bridge: Lorentzian spacetime bridge Calibration / nuisance: worldline and clock model Falsification criterion: clock ratios/redshifts fail
Energy	SI / standard unit: J or eV Experimental measurement: calorimetry/spectroscopy BFG readout target: E_{exp} , E_{bind} ; H_{hat} spectrum Required bridge: M_E and quantum bridge Calibration / nuisance: zero point, detector calibration Falsification criterion: spectral/energy differences fail
Frequency	SI / standard unit: Hz Experimental measurement: spectroscopy/clock counting BFG readout target: transition $\Delta E/h$ after bridge Required bridge: M_Q plus h convention Calibration / nuisance: line shape, Doppler/systematics Falsification criterion: predicted line centers fail
Mass	SI / standard unit: kg; eV/ c^2 Experimental measurement: inertial/gravitational or particle reconstruction BFG readout target: particle/formation mass readout Required bridge: Π_{SM} or gravitational bridge Calibration / nuisance: detector/model systematics Falsification criterion: mass predictions fail PDG/precision data
Electric charge	SI / standard unit: C or multiples of e Experimental measurement: charge/current standards, scattering BFG readout target: charge representation from Π_{SM} Required bridge: gauge/representation bridge Calibration / nuisance: detector gain, e convention Falsification criterion: charge assignments or ratios fail
Momentum	SI / standard unit: kg m s ⁻¹ ; eV/ c Experimental measurement: tracking, recoil, scattering BFG readout target: generator/kinematic readout Required bridge: spacetime + quantum bridge Calibration / nuisance: field map, alignment Falsification criterion: kinematic distributions fail
Angular momentum / spin	SI / standard unit: J s; $\hbar$ units Experimental measurement: Stern–Gerlach, spectroscopy, collider angular analyses BFG readout target: representation label/operator Required bridge: quantum + particle bridge Calibration / nuisance: analyzer orientation Falsification criterion: spin/multiplicity predictions fail
Cross section	SI / standard unit: m ² ; barn Experimental measurement: event rates with luminosity BFG readout target: scattering probability/amplitude Required bridge: quantum field bridge Calibration / nuisance: luminosity, efficiency Falsification criterion: predicted cross sections fail
Decay rate / lifetime	SI / standard unit: s ⁻¹ / s Experimental measurement: time distributions BFG readout target: transition/decay generator Required bridge: particle quantum bridge Calibration / nuisance: acceptance/background Falsification criterion: lifetime/branching predictions fail

Symbol or observable	Complete specification
Dimensionless couplings	SI / standard unit: 1 Experimental measurement: scattering/precision fits BFG readout target: $g_i(\Xi)$ Required bridge: Π_{SM} Calibration / nuisance: renormalization scale Falsification criterion: running/couplings fail independent fits
Gravitational acceleration	SI / standard unit: $m s^{-2}$ Experimental measurement: free fall, gravimetry BFG readout target: geodesic/field prediction from g Required bridge: M_g + dynamics Calibration / nuisance: local environment Falsification criterion: acceleration/redshift/lensing fail
Curvature / tidal field	SI / standard unit: m^{-2} or s^{-2} forms Experimental measurement: geodesic deviation, gravitational observations BFG readout target: Riemann/Ricci tensors from g Required bridge: connection/curvature bridge Calibration / nuisance: source model Falsification criterion: tidal/lensing/wave observables fail
Quantum outcome probability	SI / standard unit: 1 Experimental measurement: repeated POVM/instrument frequencies BFG readout target: $\text{tr}(E_a \rho_Q)$ Required bridge: Π_Q plus instrument map Calibration / nuisance: state preparation, detector tomography Falsification criterion: holdout frequencies reject predicted probabilities
Witness survival mass	SI / standard unit: 1 Experimental measurement: kernel-internal; no established direct apparatus yet BFG readout target: m_{sel}, m_4, m_{tot} Required bridge: requires physical witness identification Calibration / nuisance: preparation/selection protocol Falsification criterion: bridge rejected if measured success statistics disagree
Information I	SI / standard unit: bit/nat or 1 Experimental measurement: specified ensemble/configuration counting BFG readout target: $\log C_\theta(X) $ Required bridge: M_I fixes $X, C_\theta, \theta, \text{base}, I_0$ Calibration / nuisance: coarse graining Falsification criterion: independent counts disagree
Order parameter ϕ	SI / standard unit: 1 Experimental measurement: derived from independently measured energy/information ratios BFG readout target: ϕ_E or ϕ_N Required bridge: M_ϕ Calibration / nuisance: reference I_0 and energy definitions Falsification criterion: derived observable disagrees
Biological flux	SI / standard unit: $\text{mol s}^{-1}, \text{kg s}^{-1}, W$ as appropriate Experimental measurement: transport/metabolic assays BFG readout target: J_{in}, J_{out}, J_{net} Required bridge: Π_{bio} Calibration / nuisance: boundary and species definitions Falsification criterion: flux/viability predictions fail
World/Self functional readouts	SI / standard unit: task-dependent Experimental measurement: behavioral/interventional assays BFG readout target: Π_W, Π_S, U_{WS} Required bridge: cognitive bridge Calibration / nuisance: task model and intervention class Falsification criterion: out-of-sample causal predictions fail

Symbol or observable	Complete specification
Phenomenal report / experience	SI / standard unit: no single SI unit Experimental measurement: report, behavior, physiology under protocol BFG readout target: Γ_{subj} only after definition Required bridge: H_{phen} Calibration / nuisance: report model and confounds Falsification criterion: pre-registered bridge predictions fail

Table 4. Canonical physical-observable interface for the Balance–Field Equation. Entries marked as bridges are model commitments to be fixed before holdout testing.

41.1 Metrological requirement

The measurement map is not permitted to identify an internal variable with an SI observable merely because their dimensions can be made compatible. Each observable requires an apparatus model and a calibration chain. The SI provides the internationally agreed unit system, while precision particle properties are independently compiled by the Particle Data Group. In the quantum sector, electrical and quantum standards illustrate how physical laws become metrological realizations only after a concrete instrument and traceability chain are specified.

41.2 Minimum empirical closure condition

For a preparation D_{prep} , calibration C , bridge B , and holdout data D_{test} , a physically closed BFG sector must provide a predictive distribution $P_{\text{BFG}}(D_{\text{test}} | D_{\text{prep}}, C, B)$ before D_{test} is observed. A sector is not physically closed by notation alone. This criterion applies independently to time, geometry, quantum outcomes, particle parameters, biological observables, and subject-level observables.

42 Complete Canonical Mathematical Specification

This section consolidates the complete finite Balance–Field Equation in one place. It is intended to remove the need to reconstruct types, dimensions, assumptions, and successor rules from separate passages.

$$[\Xi_{k+1}] = \text{Ubar}_{\text{BFG}} \setminus ([\Xi_k])$$

The quotient operator factors into selection, first transport, endogenous metric projection, positive load weighting, ambient reduced SVD, successor geometry and kernel, isometric R4 realignment, formation inheritance, and the seed or terminal rule. The nominal second normalization has mass exactly one.

42.1 Domain of one nonterminal step

All input operators act on H_k . T_{sel} maps the selected source support to P_{sel} . The analysis operator is $A: H_k \rightarrow H_k \oplus H_k$, without domain restriction. Its reduced SVD has $U: C^r \rightarrow H_k \oplus H_k$ and $V: C^r \rightarrow H_k$. The successor is C^r . Positive dual loads imply $\text{rank}(P_{\text{sel}}) \leq r \leq n_k$. Padding null directions or substituting AZ changes the model.

42.2 Gauge-invariant successor

The successor is an equivalence class under simultaneous unitary coordinate changes. A common phase of a simple singular-vector pair and a unitary rotation within an exactly degenerate singular block are not distinct physical or mathematical successors. Under $U \rightarrow UW$ and $V \rightarrow VW$ with W commuting with Σ , K_+ , Y_+ , S_4^{+} , ρ_W , $+$, ρ_F , $+$ and P_+ all transform by the same target-space conjugation. Thus the quotient state $[\Xi_+]$ is unique even when a matrix representation is not.

42.3 Regular strata and differentiability

Regularity is assumed on fixed projector and matrix-rank strata: a gap to the threshold 1, positive m_{sel} and both loads, fixed positive rank of A, simple separated positive singular values for componentwise formulas, and a fixed seed branch. On $E \neq 0$ require a simple negative lowest eigenvalue separated from zero and the other eigenvalues; on $E=0$ keep complement dimension zero. Input tangent variations are Hermitian in K,F,W,Y, have $\text{tr}(\delta W)=0$, and satisfy $P \delta P P = (I-P) \delta P (I-P) = 0$. Positivity/rank constraints also restrict allowed tangents. These are specified regularity assumptions, not universally necessary conditions for every possible proof. All polar norms here are operator 2-norms.

42.4 Terminal and continuation semantics

Apply terminal gates in order: zero selection rank, zero first witness mass, failure of positive dual loads, then seed failure on $E \neq 0$. Lack of a negative minimum gives T_{seednone} ; a multiple negative minimum gives T_{seeddeg} . After positive loads, zero A/R4 rank and zero m_4 are impossible and are only implementation consistency checks. $E=0$ is the sole no-seed continuation.

42.5 Stability hierarchy

Endogenous geometry is positive and the actual ambient update additionally gives $0 < Y_+ < I$. Neither result bounds the derivative of the complete tuple by a constant below one. Regional full-kernel contraction remains open; rank, gap, load and mass margins only describe regions on which it can be investigated. The neutral K/F control branch excludes a universal strict product-norm theorem. Transverse stability requires a specified invariant set and a separate certificate.

43 Complete Variable and Operator Dictionary

Every symbol used in the canonical equation is listed below. 'Bridge' means that the finite mathematical kernel does not by itself determine the physical identification.

Symbol or observable	Complete specification
Φ_0	Domain / type: X_0 Codomain / unit: ontological Exact definition: Level-0 object Mathematical role: pre-operational domain Conditions: no canonical Ξ map Status: Definition/bridge
Ξ_k	Domain / type: product operator space Codomain / unit: mixed Exact definition: $(K, \rho_F, \rho_W, Y, P)$ Mathematical role: complete finite state Conditions: finite H_k Status: Definition
H_k	Domain / type: finite complex Hilbert space Codomain / unit: $\dim n_k$ Exact definition: active event space Mathematical role: carrier of Ξ_k Conditions: finite Status: Definition
K_k	Domain / type: $\text{Herm}(H_k)$ Codomain / unit: E_F Exact definition: $K=K^\dagger$ Mathematical role: kernel/formation geometry Conditions: Hermitian Status: Definition

Symbol or observable	Complete specification
$\rho_{F,k}$	Domain / type: Pos(H_k) Codomain / unit: E_F Exact definition: $\rho_F \geq 0$ Mathematical role: formation Conditions: not necessarily trace-one Status: Definition
$\rho_{W,k}$	Domain / type: Dens(H_k) Codomain / unit: 1 Exact definition: $\rho_W \geq 0, \text{tr}=1$ Mathematical role: witness Conditions: normalized input Status: Definition
Y_k	Domain / type: Herm_+(H_k) Codomain / unit: 1 Exact definition: $Y \geq 0$ Mathematical role: Gram/load geometry Conditions: positive semidefinite Status: Definition
P_k	Domain / type: Proj(H_k) Codomain / unit: 1 Exact definition: $P = P^\dagger = P^2$ Mathematical role: persistent carrier Conditions: orthogonal projector Status: Definition
Q_k	Domain / type: Proj(H_k) Codomain / unit: 1 Exact definition: $1_{[0,1)}(Y)$ Mathematical role: spectral selection Conditions: gap to 1 for smoothness Status: Definition
C_k	Domain / type: End(H_k) Codomain / unit: 1 Exact definition: $(I+Y)^{-1}$ Mathematical role: neutral/complement channel Conditions: $Y \geq 0$ Status: Definition
B_k	Domain / type: End(H_k) Codomain / unit: 1 Exact definition: $YC = I - C$ Mathematical role: balance channel Conditions: $Y \geq 0$ Status: Definition
G_k	Domain / type: PosDef(H_k) Codomain / unit: 1 Exact definition: $I+Y$ Mathematical role: internal metric Conditions: $Y \geq 0$ Status: Theorem/definition
I and I_0	Domain / type: specified configuration count and reference Codomain / unit: bit/nat or dimensionless Exact definition: $I = \log_b C_\theta(X) $ Mathematical role: information readout Conditions: finite nonempty class, $b > 1, I_0 > 0$ Status: Open bridge

Symbol or observable	Complete specification
φ_N or φ_E	Domain / type: scalar ratios Codomain / unit: dimensionless Exact definition: $(N_+/N_-)(I/I_0)$ or $(E_{\text{exp}}/E_{\text{bind}})(I/I_0)$ Mathematical role: order readout Conditions: positive references; distinct from Φ_0 Status: Open bridge

Table 5. Complete Level-0 and finite-state variable dictionary.

Symbol or observable	Complete specification
X_{sel}	Domain / type: $\text{End}(H_k)$ Codomain / unit: 1 Exact definition: $Q_k P_k$ Mathematical role: selection overlap Conditions: compatible source space Status: Definition
T_{sel}	Domain / type: partial isometry Codomain / unit: 1 Exact definition: $\text{polar}(X_{\text{sel}})$ Mathematical role: first transport Conditions: $\text{rank} > 0$ Status: Definition
$S_{\text{sel}}^\wedge$	Domain / type: projector Codomain / unit: 1 Exact definition: $T_{\text{sel}}^\dagger T_{\text{sel}}$ Mathematical role: source support Conditions: T_{sel} defined Status: Definition
P_{sel}	Domain / type: projector Codomain / unit: 1 Exact definition: $T_{\text{sel}} T_{\text{sel}}^\dagger$ Mathematical role: target support Conditions: T_{sel} defined Status: Definition
ρ_F^{sel}	Domain / type: positive operator Codomain / unit: E_F Exact definition: $T_{\text{sel}} \rho_F T_{\text{sel}}^\dagger$ Mathematical role: selected formation Conditions: transport defined Status: Definition
$\rho_{\text{un}}^{\text{sel}}$	Domain / type: positive operator Codomain / unit: 1 Exact definition: $T_{\text{sel}} \rho_W T_{\text{sel}}^\dagger$ Mathematical role: unnormalized witness Conditions: transport defined Status: Definition
m_{sel}	Domain / type: $[0,1]$ Codomain / unit: 1 Exact definition: $\text{tr}(S_{\text{sel}}^\wedge \rho_W)$ Mathematical role: first survival mass Conditions: $\text{tr} \rho_W = 1$ Status: Theorem/definition
ρ_W^{sel}	Domain / type: density operator Codomain / unit: 1 Exact definition: $\rho_{\text{un}}^{\text{sel}}/m_{\text{sel}}$ Mathematical role: conditioned witness Conditions: $m_{\text{sel}} > 0$ Status: Definition

Symbol or observable	Complete specification
Z	Domain / type: isometry Codomain / unit: 1 Exact definition: $Z^\dagger Z = I, ZZ^\dagger = P_{\text{sel}}$ Mathematical role: target-support frame Conditions: local gauge Status: Definition
P_G	Domain / type: G-projector Codomain / unit: 1 Exact definition: $Z(Z^\dagger GZ)^{-1}Z^\dagger G$ Mathematical role: metric projection Conditions: invertible compression Status: Definition
Λ_C	Domain / type: R_+ Codomain / unit: load unit Exact definition: trace formula Mathematical role: C-load Conditions: positive branch Status: Definition
Λ_B	Domain / type: R_+ Codomain / unit: load unit Exact definition: trace formula Mathematical role: B-load Conditions: positive branch Status: Definition
α	Domain / type: (0,1) Codomain / unit: 1 Exact definition: $\Lambda_B / (\Lambda_C + \Lambda_B)$ Mathematical role: C weight Conditions: dual-load branch Status: Definition
β	Domain / type: (0,1) Codomain / unit: 1 Exact definition: $\Lambda_C / (\Lambda_C + \Lambda_B)$ Mathematical role: B weight Conditions: dual-load branch Status: Definition

Table 6. Selection, transport, metric-projection, and load variables.

Symbol or observable	Complete specification
A_k	Domain / type: $H_k \rightarrow H_k \oplus H_k$ Codomain / unit: analysis space Exact definition: vertical stack of weighted P_{GC}/P_{GB} Mathematical role: analysis operator Conditions: positive loads Status: Definition
r_k	Domain / type: N Codomain / unit: 1 Exact definition: $\text{rank}(A_k)$ Mathematical role: successor dimension Conditions: $r_k \geq 1$ Status: Definition
U_k	Domain / type: isometry $C^r \rightarrow$ analysis Codomain / unit: 1 Exact definition: left reduced SVD frame Mathematical role: successor kernel frame Conditions: modulo gauge Status: Definition

Symbol or observable	Complete specification
Σ_k	Domain / type: positive diagonal Codomain / unit: 1 Exact definition: singular spectrum Mathematical role: sets Y_+ Conditions: positive singular values Status: Definition
V_k	Domain / type: isometry C^r -source Codomain / unit: 1 Exact definition: right reduced SVD frame Mathematical role: R_4 source frame Conditions: modulo gauge Status: Definition
Ω_U	Domain / type: skew-Hermitian Codomain / unit: variation Exact definition: $U^\dagger \delta U$ Mathematical role: SVD connection Conditions: simple stratum Status: Derived
Ω_V	Domain / type: skew-Hermitian Codomain / unit: variation Exact definition: $V^\dagger \delta V$ Mathematical role: SVD connection Conditions: simple stratum Status: Derived
K_+	Domain / type: $\text{Herm}(C^r)$ Codomain / unit: E_F Exact definition: $U^\dagger(K \oplus K)U$ Mathematical role: successor kernel Conditions: K Hermitian Status: Definition
Y_+	Domain / type: $\text{Herm}_+(C^r)$ Codomain / unit: 1 Exact definition: Σ^2 Mathematical role: successor Gram geometry Conditions: reduced SVD Status: Theorem/definition
X_{R_4}	Domain / type: linear map Codomain / unit: 1 Exact definition: $V^\dagger P_{\text{sel}}$ Mathematical role: R_4 overlap Conditions: compatible frames Status: Definition
T_4	Domain / type: partial isometry Codomain / unit: 1 Exact definition: $\text{polar}(X_{R_4})$ Mathematical role: R_4 transport Conditions: $\text{rank} > 0$ Status: Definition
S_4^-	Domain / type: projector Codomain / unit: 1 Exact definition: $T_4^\dagger T_4$ Mathematical role: R_4 source support Conditions: T_4 defined Status: Definition

Symbol or observable	Complete specification
S_4^+	Domain / type: projector Codomain / unit: 1 Exact definition: $T_4 T_4^\dagger$ Mathematical role: R4 target support Conditions: T_4 defined Status: Definition
m_4	Domain / type: {1} Codomain / unit: 1 Exact definition: $\text{tr}(S_4^+ \rho_W^{\text{sel}})$ Mathematical role: second survival mass Conditions: identically one on positive dual-load branch Status: Definition/theorem
m_{tot}	Domain / type: [0,1] Codomain / unit: 1 Exact definition: $m_{\text{sel}} m_4$ Mathematical role: step survival mass Conditions: sequential conditioning Status: Definition

Table 7. Reduced-SVD, successor, and R4 variables.

Symbol or observable	Complete specification
E	Domain / type: projector Codomain / unit: 1 Exact definition: $I - S_4^+$ Mathematical role: free target complement Conditions: successor space fixed Status: Definition
Z_E	Domain / type: isometry Codomain / unit: 1 Exact definition: $Z_E Z_E^\dagger = E$ Mathematical role: complement frame Conditions: horizontal gauge locally Status: Definition
H_E	Domain / type: $\text{Herm}(\text{Ran } E)$ Codomain / unit: E_F Exact definition: $Z_E^\dagger K_+ Z_E$ Mathematical role: seed spectral operator Conditions: K_+ Hermitian Status: Definition
λ	Domain / type: $\mathbb{R}$ Codomain / unit: E_F Exact definition: simple negative minimum of H_E Mathematical role: seed amplitude Conditions: $\lambda < 0$ Status: Definition
π	Domain / type: rank-one projector Codomain / unit: 1 Exact definition: $vv^\dagger$ Mathematical role: reduced seed support Conditions: simple minimum Status: Definition
P_{seed}	Domain / type: projector Codomain / unit: 1 Exact definition: $Z_E \pi Z_E^\dagger$ Mathematical role: embedded seed support Conditions: $P_{\text{seed}} \leq E$ Status: Definition

Symbol or observable	Complete specification
κ_F	Domain / type: R_+ Codomain / unit: conversion Exact definition: fixed unit factor Mathematical role: K-to-formation conversion Conditions: constant Status: Assumption if needed
ρ_{seed}	Domain / type: positive operator Codomain / unit: E_F Exact definition: $-\kappa_F \lambda \Pi_{\text{seed}}$ Mathematical role: new formation Conditions: negative λ Status: Definition
$\rho_{F,+}$	Domain / type: positive operator Codomain / unit: E_F Exact definition: $\rho_{F,\text{inh}} + \rho_{\text{seed}}$ Mathematical role: successor formation Conditions: $E=0$ or simple negative seed; otherwise terminal Status: Definition
P_+	Domain / type: projector Codomain / unit: 1 Exact definition: $S_4^{++} \Pi_{\text{seed}}$ Mathematical role: successor carrier Conditions: orthogonal supports Status: Theorem/definition
R_λ	Domain / type: reduced resolvent Codomain / unit: E_F^{-1} Exact definition: $(H_E - \lambda I)^{+}$ Mathematical role: seed differential Conditions: simple eigenvalue Status: Derived

Table 8. Seed, complement, and formation variables.

Symbol or observable	Complete specification
Ψ	Domain / type: 9-role state Codomain / unit: mixed Exact definition: $(\varphi_{\text{fun}}, I_{\text{fun}}, A_{\text{fun}}, G_{\text{fun}}, \Omega_{\text{fun}}, T_{\text{fun}}, S_{\text{fun}}, M_{\text{fun}}, C_{\text{fun}})$ Mathematical role: functional readout Conditions: Π incomplete Status: Bridge
Π	Domain / type: map Codomain / unit: - Exact definition: $\Xi \dashv \Psi$ Mathematical role: factor projection Conditions: fiber condition Status: Bridge
T_{clock}	Domain / type: $X_\Xi / \text{Unitary} \rightarrow R_{\{>0\}}$ Codomain / unit: s Exact definition: physical step duration Mathematical role: event-to-time bridge Conditions: calibration Status: Bridge
t_k	Domain / type: R Codomain / unit: s Exact definition: $t_0 + \sum T_{\text{clock}}$ Mathematical role: physical time Conditions: clock fixed Status: Bridge

Symbol or observable	Complete specification
$g_{\mu\nu}$	Domain / type: Lorentz metric Codomain / unit: 1 Exact definition: bridge-dependent Mathematical role: spacetime Conditions: manifold/signature Status: Bridge
ρ_Q	Domain / type: density operator Codomain / unit: 1 Exact definition: ρ_W or ρ_F /tr ρ_F candidates Mathematical role: quantum state Conditions: experiment selects bridge Status: Bridge
H_{hat}	Domain / type: self-adjoint operator Codomain / unit: J/eV Exact definition: quantum Hamiltonian Mathematical role: energy generator Conditions: Π_Q Status: Bridge
E_a	Domain / type: POVM effect Codomain / unit: 1 Exact definition: $0 \leq E_a \leq I$ Mathematical role: quantum measurement Conditions: instrument fixed Status: External standard
Π_{SM}	Domain / type: map Codomain / unit: - Exact definition: $\Xi \rightarrow$ particle/gauge data Mathematical role: particle bridge Conditions: representation/couplings Status: Bridge
$J_{\text{in/out/net}}$	Domain / type: fluxes Codomain / unit: system-dependent Exact definition: $J_{\text{net}} = J_{\text{out}} - J_{\text{in}}$ Mathematical role: biological transport Conditions: boundary fixed Status: Bridge/readout
W	Domain / type: functional state Codomain / unit: task-dependent Exact definition: $\Pi_W(\text{history})$ Mathematical role: world model Conditions: readout defined Status: Bridge
S	Domain / type: functional state Codomain / unit: task-dependent Exact definition: $\Pi_S(\text{history})$ Mathematical role: self model Conditions: readout defined Status: Bridge
Γ_{subj}	Domain / type: gate/readout Codomain / unit: 1 Exact definition: future explicit function Mathematical role: subject criterion Conditions: measurement map Status: Bridge

Table 9. Functional, physical, biological, and subject-level bridge variables.

44 Reader's Guide to the Recursive Mechanism

One recursive event begins with the declared five operators. Q selects the subthreshold spectral sector and QP defines the first directed polar transport. Formation retains its unnormalized magnitude; witness is conditioned and its initial mass is separately recorded. G and P_G evaluate formation through complementary C and B channels, whose loads determine the weights. The full ambient A supplies the active successor rank, geometry and frames. R_4 realigns the complete selected carrier without a second loss. A nonempty target complement requires a unique negative lowest seed mode. The whole event therefore specifies every intermediate type, transport, normalizing factor and terminal test.

45 Detailed Selection Logic

Selection is not deletion by a scalar threshold alone. $Q_k = 1_{-[0,1)}(Y_k)$ is a spectral projector, so it selects invariant spectral subspaces of the Hermitian positive operator Y_k . P_k is an independent persistence projector. The product $Q_k P_k$ need not itself be a projector, which is why its polar factor is used. $T_{sel,k}$ retains the geometric relation between the selected Y -subspace and the persistent P -subspace. Its source projector $S_{sel}^\wedge$ and target projector P_{sel} therefore carry different information. This distinction is also what makes the first witness mass $m_{sel} = \text{tr}(S_{sel}^\wedge \cdot \rho_w)$ mathematically meaningful: the mass is evaluated on the source before the state is transported and renormalized.

46 Why the Metric Projector Is Endogenous

P_G is the G -orthogonal projector onto $\text{Ran}(P_{sel})$. It equals the ordinary Euclidean projector P_{sel} exactly when $[P_{sel}, G] = 0$; G need not equal I . It satisfies $P_G^2 = P_G$ and $P_G^\dagger G = G P_G$. The current geometry affects selection, projection, loads and ultimately its own successor through A . This self feedback is explicit; its stabilizing effect is proved on the scalar branch and bounded generally by $0 < Y_+ < I$, without a claim of universal full-state contraction.

47 Load Weighting and Neutral Balance

$C = (I + Y)^{-1}$ and $B = Y(I + Y)^{-1}$ satisfy $C + B = I$ and commute with Y . In the eigenbasis of Y , an eigenvalue $y \geq 0$ gives $c = 1/(1+y)$ and $b = y/(1+y)$. Hence low-load directions are C -dominated, while high-load directions are B -dominated. The global weights α and β are not these eigenvalue-wise coefficients; they are formed from the metric formation loads Λ_C and Λ_B . The cross-definition $\alpha = \Lambda_B / (\Lambda_C + \Lambda_B)$, $\beta = \Lambda_C / (\Lambda_C + \Lambda_B)$ makes each analysis branch sensitive to the strength of the complementary load. This is the finite-kernel realization of a balance-mediated analysis rather than a one-channel amplification rule.

48 Reduced SVD as the Dimension-Formation Step

The reduced SVD retains precisely the positive singular directions of the ambient A . Thus $n_+ = \text{rank}(A) \leq n$. Although r can exceed the selected-carrier rank, the finite ambient dimension cannot grow. Rank-loss boundaries change the active state space. Fixed-rank derivatives require the declared gaps; coordinate gauge and physical emergence are different issues.

49 Gauge Freedom and Physical Equivalence

An SVD is never a unique ordered list of vectors in an absolute sense. A simple singular pair can be multiplied by a common complex phase; a degenerate singular block can be rotated by a unitary matrix. If K_+ , Y_+ , R_4 , formation, and witness were read as absolute matrices without accounting for this freedom, the update would be representation-dependent. The canonical equation therefore defines the successor on a unitary quotient. Every target-space object is transformed by the same conjugation. Observable scalar invariants such as traces and spectra are unchanged, while coordinate matrices change covariantly. This is a mathematical gauge

principle internal to the finite recursion; it should not be confused with the Standard Model gauge group, which is a separate physical bridge target.

50 Polar Differential Interpretation of the Three Terms

The constant-rank polar derivative $\delta T = T \Omega + (I - S_-) E_{X_H}^+ + T H^+ + T^\dagger E_X (I - S_-)$ has three geometrically distinct pieces. $T \Omega$ rotates coordinates inside the active source support. The second term moves the target support transversely. The third term moves the source support and transports that motion through T . Thus support motion is not an independent perturbation; it is determined by $E_X = \delta X$. The $3 \|E_X\| / \sigma_k$ bound is deliberately conservative. Its dependence on the smallest positive singular value expresses a standard fact of rank-sensitive matrix geometry: near rank loss, the active support can become highly sensitive even when a particular path remains benign.

51 R4 as Recursive Re-Alignment

Because $\text{Ran}(P_{\text{sel}})$ lies in $\text{Ran}(A^\dagger)$, $V V^\dagger P_{\text{sel}} = P_{\text{sel}}$. Thus $X_{R4}^\dagger X_{R4} = P_{\text{sel}}$ and $T_4 = V^\dagger P_{\text{sel}}$ already is a partial isometry. $R4$ realigns the complete selected carrier, gives $m_4=1$ and determines S_4^+ and its complement. It is not a second lossy test in this class. A physical Kraus interpretation would require an independent instrument bridge.

52 Formation, Witness, and the Meaning of Seed

Inherited formation keeps its trace, whereas witness is normalized and original mass is kept in the separate record w . In this kernel $m_4=1$, so $R4$ discards no mass. For $E=0$ no seed is needed. For $E \neq 0$ a simple negative minimum gives positive seed formation; no negative minimum or a multiple negative minimum is terminal. Orthogonality preserves the projector sum. Neither seed nor complement is automatically physical particle creation.

53 Terminality as Part of the Equation

Terminality is part of the definition. Selection failure, first witness failure, or nonpositive dual loads terminate before analysis. After positive loads A and $R4$ have nonzero rank and $m_4=1$, so second-mass failure is unreachable. If $E \neq 0$ the unique rank-one seed requires a simple strictly negative minimum; its absence or degeneracy terminates. $E=0$ alone continues without a seed.

54 What the Scalar Damping Result Does and Does Not Prove

The exact scalar map $f(y) = 2y^2 / [(1+y)^2(1+y^2)]$ provides a rare closed-form control branch of the finite recursion. Algebraically $f(y) < y$ for every $y > 0$, while the actual nonterminal selection-first realization under discussion is restricted to $0 < y < 1$. The result proves monotone damping of that scalar geometric load on the stated branch. It does not prove that the complete tuple contracts because K and formation can remain neutral. This is why the appropriate general stability target is likely transverse: perturbations away from a persistent manifold may decay while coordinates along the persistent identity remain neutral.

55 From Recursive Events to Measured Time

The event index k orders recursive updates and nothing more. To compare the equation with clocks, one needs $T_{\text{clock}}(\Xi)$, a positive duration assigned to each event. Summing these durations produces physical time along a trajectory. If a continuous effective state Ψ is desired, a separate flow F_Ψ must exist such that one discrete BFG step equals the flow evaluated at the assigned duration. The clock can vary even when the projected state is unchanged; only a clock required to be a function of Ψ alone must be constant on Π -fibers. This distinction prevents event count, model time, physical seconds, and renormalization scale from being conflated.

56 From Positive Gram Geometry to Lorentzian Spacetime

The internal $G=I+Y$ is positive definite because its role is to measure directions in the finite analysis space. Spacetime requires an indefinite Lorentzian metric. Therefore a physical geometry bridge must change mathematical type. A disformal ansatz can produce a Lorentzian metric only after a manifold, background η , scalar field, and nondegeneracy condition are supplied. An Einstein-scalar action can then generate field equations, but those equations follow from the added action. The finite BFG becomes a gravitational theory only when a map from recursive states to the spacetime fields and their calibrated constants is fixed and tested.

57 Quantum Measurement as a Discriminating Bridge

The two candidate identifications $\rho_Q=\rho_W$ and $\rho_Q=\rho_F/\text{tr}(\rho_F)$ are useful precisely because they are not equivalent. In the explicit two-dimensional control state they predict 3/4 and 1/4 for the same externally fixed effect. This turns a vague analogy between kernel matrices and density operators into a falsifiable bridge choice. To perform the test physically, the preparation must determine the relevant BFG state independently of the holdout outcomes, the effect must be calibrated, and the statistical rejection rule must be fixed in advance. The R4 success effect is not substituted for the external measurement unless instrument compatibility has separately been established.

58 Particle, Atomic, Molecular, and Biological Ordering

The hierarchy is constrained by dependence. A particle bridge must precede a quantitative atom bridge because atomic Hamiltonians require masses, charges, couplings, and quantum dynamics. A molecular bridge requires atomic degrees of freedom and interactions. A biological closure bridge then requires molecular transport, boundaries, energy/material fluxes, repair, and viable persistence. The notation Cl_N is therefore treated as a placeholder for a future typed closure operation, not as proof that a named level follows from the previous one. The ordering is a research dependency graph.

59 Witness to Subject Structure

The witness is already an internal mathematical object; world and self are not. A subject-level theory therefore cannot be obtained merely by renaming ρ_W . Π_W and Π_S must construct world- and self-related functional states from the physical/history state, and U_{WS} must close their dynamics. A candidate Γ_{subj} must then specify measurable criteria such as predictive sufficiency, self-indexing, directed lineage, and admissibility. Even if these conditions are operationally satisfied, the final equivalence with phenomenal experience remains an empirical-ontological bridge rather than a theorem of matrix recursion.

60 How the Physical Measurement Layer Closes a Sector

For each physical sector, the measurement interface has four layers: preparation, state identification, prediction, and comparison. Preparation produces D_{prep} . A fixed map M_{prep} identifies Ξ or the relevant quotient class. A bridge B maps the state to the physical observable model and generates a predictive distribution. An independently calibrated apparatus produces D_{test} . The bridge is rejected or retained according to a pre-registered uncertainty model and criterion. This architecture is what turns a mathematically closed recursive kernel into a potentially physical world equation.

61 Canonical Theorem and Assumption Register

Theorem A — Ambient Gram bound: positive dual loads imply $0<Y_+<I$ and $n_+\leq n$; $G=I+Y>0$ for the declared input class.

Theorem B — Witness transport: m_{sel} lies in $[0,1]$, $m_4=1$, $m_{\text{tot}}=m_{\text{sel}}$; later nonterminal events preserve original witness mass.

Theorem C — Gauge well-definedness: simultaneous unitary changes inside allowed SVD gauge blocks conjugate every target-space successor object and therefore leave $[\Xi_+]$ unchanged.

Theorem D — Scalar damping: $f(y)<y$ for $y>0$; the checked selection-first branch uses $0<y<1$.

Theorem E — Positive fixed-point stability of the scalar ODE: negative trace and positive determinant imply local asymptotic stability on the declared positive branch.

Assumption A — Level-0 to finite-state formation is not fixed by the finite kernel.

Assumption B — Physical time requires T_{clock} .

Assumption C — Lorentzian geometry requires a physical geometry bridge.

Assumption D — Quantum state identification requires Π_Q ; ρ_W and normalized ρ_F are competing hypotheses.

Assumption E — Standard-Model parameters require Π_{SM} and cannot be inferred from gauge labels alone.

Assumption F — World/Self and phenomenality require additional functional and empirical bridges.

62 Full Differential Chain Selection to the Analysis Operator

62.1 Spectral projector variation

On a fixed selection stratum the spectrum of Y remains separated from the threshold 1. The spectral projector $Q=1_{[0,1)}(Y)$ is then differentiable as an invariant-subspace projector. In a contour representation enclosing the selected spectrum but not the excluded spectrum, δQ is determined by the resolvent of Y and δY . The important structural point is that δQ is not an independent input: it is a linear function of δY on the fixed-gap stratum. At a threshold crossing this derivative description ceases to be the correct model and the system changes stratum.

62.2 Selection overlap and polar transport

The selection overlap is $X_{\text{sel}}=QP$. Its variation is $\delta X_{\text{sel}}=(\delta Q)P+Q(\delta P)$. The constant-rank polar theorem then determines δT_{sel} from δX_{sel} , including both source- and target-support motion. Consequently $\delta S_{\text{sel}}^\wedge$ and δP_{sel} are determined by the original five input variations; no additional support perturbation is introduced.

62.3 Selected formation and witness

The transported formation obeys $\delta \rho_F^{\text{sel}}=(\delta T_{\text{sel}})\rho_F T_{\text{sel}}^\dagger + T_{\text{sel}}(\delta \rho_F)T_{\text{sel}}^\dagger + T_{\text{sel}}\rho_F(\delta T_{\text{sel}})^\dagger$. The unnormalized selected witness obeys the identical three-term transport rule with ρ_W . The first survival mass varies as $\delta m_{\text{sel}}=\text{tr}[(\delta S_{\text{sel}}^\wedge)\rho_W + S_{\text{sel}}^\wedge(\delta \rho_W)]$. The normalized selected witness therefore contains both direct transport variation and the normalization correction $-\rho_W^\wedge \delta m_{\text{sel}}/m_{\text{sel}}^2$.

62.4 Neutral split and Gram metric

Because $C=(I+Y)^\wedge$, $\delta C=-C(\delta Y)C$. Since $B=I-C$, $\delta B=-\delta C=C(\delta Y)C$. $G=I+Y$ gives $\delta G=\delta Y$. These identities make the neutral/balance variations fully endogenous to δY .

62.5 Horizontal support frame

Choose an orthonormal support frame Z with $ZZ^\dagger=P_{\text{sel}}$ and horizontal variation $\delta Z=(\delta P_{\text{sel}})Z$. Let $H_Z=Z^\dagger GZ$. Then $\delta H_Z=(\delta Z)^\dagger GZ + Z^\dagger(\delta G)Z + Z^\dagger G(\delta Z)$, and $\delta H_Z^\wedge = -H_Z^\wedge(\delta H_Z)H_Z^\wedge$. Therefore every term in δP_G follows by the product rule from δP_{sel} and δY .

62.6 Load derivatives

For $\Lambda_C = \text{tr}(G P_G C \rho_F^{\text{sel}} C P_G^\dagger)$, $\delta \Lambda_C$ is the sum of the variations of its six nonconstant factors. Explicitly it contains δG , δP_G on both sides, two δC terms, and $\delta \rho_F^{\text{sel}}$. Λ_B has the same structure with B . Hence $\delta \alpha$ and $\delta \beta$ are ordinary quotient-rule expressions in $\delta \Lambda_C$ and $\delta \Lambda_B$.

62.7 Analysis derivative

Write $A_C = \sqrt{\alpha} P_{GC}$ and $A_B = \sqrt{\beta} P_{GB}$. Then $\delta A_C = [\delta \alpha / (2\sqrt{\alpha})] P_{GC} + \sqrt{\alpha} (\delta P_G) C + \sqrt{\alpha} P_G (\delta C)$, and analogously for A_B . Stacking the two blocks gives δA . Thus the full analysis derivative is a closed function of the five original input variations on the declared regular stratum.

63 SVD Differential in the Complex Case

63.1 Tangent decomposition

Let $F = \delta A$ and $M = U^\dagger F V$. Define $\Omega_U = U^\dagger \delta U$ and $\Omega_V = V^\dagger \delta V$. Orthonormality implies both are skew-Hermitian. Differentiating $A = U \Sigma V^\dagger$ and compressing between $U^\dagger$ and V gives $M = \Omega_U \Sigma + \delta \Sigma - \Sigma \Omega_V$.

63.2 Diagonal phase equation

The diagonal entry is $M_{ii} = \delta \sigma_i + \sigma_i [(\Omega_U)_{ii} - (\Omega_V)_{ii}]$. Since the diagonal entries of a skew-Hermitian matrix are purely imaginary, $\delta \sigma_i = \text{Re } M_{ii}$ and $(\Omega_U)_{ii} - (\Omega_V)_{ii} = i \text{Im}(M_{ii}) / \sigma_i$. Only the common phase remains free. This is why imposing both diagonal connections equal to zero is generally inconsistent.

63.3 Off-diagonal equations

For $i \neq j$ the pair M_{ij} and conjugate M_{ji} gives a two-by-two linear system for $(\Omega_U)_{ij}$ and $(\Omega_V)_{ij}$. Its determinant is $\sigma_j^2 - \sigma_i^2$. Therefore the componentwise frame derivative requires separated singular values. The resulting coefficients become large as a singular-value gap closes, while the invariant cluster subspace can remain smooth.

For $i \neq j$, $(\Omega_U)_{ij} = (\sigma_j M_{ij} + \sigma_i \text{conjugate}(M_{ji})) / (\sigma_j^2 - \sigma_i^2)$, and $(\Omega_V)_{ij} = (\sigma_i M_{ij} + \sigma_j \text{conjugate}(M_{ji})) / (\sigma_j^2 - \sigma_i^2)$. These coefficients solve the declared complex tangent system on a simple singular-value stratum.

63.4 Normal support components

The parts of δU and δV outside their current ranges are fixed by F and Σ^{-1} : $(I - UU^\dagger) \delta U = (I - UU^\dagger) F V \Sigma^{-1}$ and $(I - VV^\dagger) \delta V = (I - VV^\dagger) F^\dagger U \Sigma^{-1}$. These are essential because R_4 depends directly on V and K_+ depends directly on U .

63.5 Successor derivatives

$\delta Y_+ = \delta \Sigma \Sigma + \Sigma \delta \Sigma$. For $K_+ = U^\dagger (K \oplus K) U$, $\delta K_+ = (\delta U)^\dagger (K \oplus K) U + U^\dagger (\delta K \oplus \delta K) U + U^\dagger (K \oplus K) \delta U$. Hence the successor derivative contains both direct K variation and the endogenous frame response.

64 R4, Witness, and Formation Differential

64.1 R4 overlap

$X_{R4} = V^\dagger P_{\text{sel}}$ gives $\delta X_{R4} = (\delta V)^\dagger P_{\text{sel}} + V^\dagger \delta P_{\text{sel}}$. The constant-rank polar derivative determines δT_4 , δS_4^- and δS_4^+ entirely from this overlap variation.

64.2 Second survival mass

The general product rule is $\delta m_4 = \text{tr}[(\delta S_4^{\wedge-})\rho_W^{\text{sel}} + S_4^{\wedge-}\delta\rho_W^{\text{sel}}]$. Within this ambient kernel it simplifies to $\delta m_4 = 0$, because $S_4^{\wedge-} = P_{\text{sel}}$ and the selected density is supported and normalized there. This does not set δT_4 to zero.

64.3 Successor witness

The three product-rule terms in $\delta(T_4 \rho_W^{\text{sel}} T_4^\dagger)$ remain necessary. Here $m_4 = 1$ and $\delta m_4 = 0$, so no second normalization correction survives. Direct frame and carrier variations still change the successor witness.

64.4 Inherited formation

$\delta \rho_{F,\text{inh}} = (\delta T_4) \rho_F^{\text{sel}} T_4^\dagger + T_4 (\delta \rho_F^{\text{sel}}) T_4^\dagger + T_4 \rho_F^{\text{sel}} (\delta T_4)^\dagger$. Formation has no witness-style normalization term.

65 Seed Eigenprojector Derivation

65.1 Moving complement

$E = I - S_4^{\wedge+}$ gives $\delta E = -\delta S_4^{\wedge+}$. Choose Z_E with $Z_{EZE}^\dagger = E$ and horizontal $\delta Z_E = (\delta E)Z_E$. The reduced Hermitian seed operator is $H_E = Z_E^\dagger K_- + Z_E$.

65.2 Reduced operator variation

Product differentiation gives $\delta H_E = (\delta Z_E)^\dagger K_- + Z_E + Z_E^\dagger \delta K_- + Z_E + Z_E^\dagger K_- + \delta Z_E$. With the horizontal frame this becomes the embedded formula already stated in the canonical section.

65.3 Simple eigenvalue

For $H_E v = \lambda v$, $\|v\| = 1$, $\pi = vv^\dagger$, first-order perturbation theory gives $\delta \lambda = v^\dagger \delta H_E v = \text{tr}(\pi \delta H_E)$. Define R_λ as the inverse of $H_E - \lambda I$ on v -perp and zero on $\text{span}(v)$, equivalently the Moore–Penrose pseudoinverse for a simple eigenvalue.

65.4 Projector derivative

Differentiating the eigenvalue equation and fixing the phase horizontally gives $\delta \pi = -R_\lambda (\delta H_E) \pi - \pi (\delta H_E) R_\lambda$. Embedding into the moving complement adds the two outer frame terms: $\delta \Pi_{\text{seed}} = (\delta E) \Pi_{\text{seed}} + \Pi_{\text{seed}} (\delta E) + Z_E (\delta \pi) Z_E^\dagger$.

65.5 Seed derivative

$\rho_{\text{seed}} = -\kappa_F \lambda \Pi_{\text{seed}}$ therefore gives $\delta \rho_{\text{seed}} = -\kappa_F [(\delta \lambda) \Pi_{\text{seed}} + \lambda \delta \Pi_{\text{seed}}]$. This closes the seed derivative on the simple negative eigenvalue stratum.

66 Algebraic Admissibility Checks

66.1 Projector property

Because $S_4^{\wedge+}$ and Π_{seed} are orthogonal projectors with $\Pi_{\text{seed}} \leq I - S_4^{\wedge+}$, their product vanishes in both orders. Hence $(S_4^{\wedge+} + \Pi_{\text{seed}})^2 = S_4^{\wedge+} + \Pi_{\text{seed}} = P_-$.

66.2 Positivity of formation

Congruence by T_4 preserves positivity, so $\rho_{F,inh} \geq 0$ whenever $\rho_F^{sel} \geq 0$. If $\lambda < 0$ and $\kappa_F > 0$, $-\kappa_F \lambda \Pi_{seed} \geq 0$. Therefore $\rho_{F,+} = \rho_{F,inh} + \rho_{seed} \geq 0$.

66.3 Witness trace

$\text{tr}(T_4 \rho_W^{sel} T_4^\dagger) = \text{tr}(S_4^\wedge - \rho_W^{sel}) = m_4$. Dividing by m_4 gives $\text{tr} \rho_{W,+} = 1$ whenever $m_4 > 0$.

66.4 Hermiticity

Hermiticity of K is preserved by $K_+ = U^\dagger(K \oplus K)U$. Hermiticity and positivity of Y_+ follow from Σ^2 . Thus the successor remains in the canonical state type on every nonterminal branch.

67 Detailed Scalar Control Branch

67.1 Setup

On the full-persistence scalar branch $P=W=1$, $F=f_0 > 0$ and $0 < y < 1$. The neutral split gives $\alpha = y^2/(1+y^2)$ and $\beta = 1/(1+y^2)$. The reduced analysis operator has one active column; U is its normalized two-component frame, $V=1$, R_4 is the identity, and the seed complement vanishes.

67.2 Successor

Direct substitution yields $K_+ = K$, $F_+ = F$, $W_+ = 1$ and $P_+ = 1$. The Gram successor is $f(y) = 2y^2/[(1+y)^2(1+y^2)]$.

67.3 Damping proof

For $y > 0$, $f(y)/y = 2y/[(1+y)^2(1+y^2)]$. The denominator minus the numerator equals $(1+y)^2(1+y^2) - 2y = 1 + 2y^2 + 2y^3 + y^4 > 0$. Hence $f(y)/y < 1$ and $f(y) < y$. Positivity is immediate.

67.4 Consequence

The branch demonstrates genuine geometric damping while K and F are neutral. Any theorem claiming strict contraction of the complete state in a fixed positive product norm would contradict these neutral directions. This control branch therefore fixes the correct scope of future stability claims.

The scalar sequence decreases toward zero. A positive limit would satisfy $f(y) = y$, which the strict inequality excludes. Consequently positive load margins cannot be uniform along its infinite tail. Geometric damping alone is not a positive stability reserve against every disturbance.

Numerical controls. Repeated double-precision checks with NumPy and random seed 20261007 used 40 complex matrix preparations. The maximum $\|S_4^\wedge - P_{sel}\|$ was $2.47e-15$, maximum $|m_{4-1}|$ was $1.67e-15$ and maximum gauge residual was $2.18e-14$. These are checks of the algebraic transport stage, not empirical validation or an assertion that every preparation passes the final seed gate.

Independent central differences gave residuals $1.70e-9$ for the polar differential, $1.10e-10$ for the simple seed projector and $1.02e-10$ for the full endogenous analysis derivative. Complex SVD tangent reconstruction had residual $2.17e-15$. Scalar damping was checked on 1000 points in $(0,1)$. Finite numerical tolerances do not replace the analytical proofs.

The graphical calculations use 31-by-31 uniform parameter grids. Ambient panels use $Y = \text{diag}(y1, 0.72)$, $P=F=W=vv^\dagger$, $v = (\cos \theta, \sin \theta)$, $0.08 \leq y1 \leq 0.92$ and $0.12 \leq \theta \leq 1.45$. Formation units are set to one. Other panels state their distinct formulas and ranges. Exactly zero singular values are not retained; numerical rank checks use a $1e-12$ threshold in the control family. The theoretical definition uses exact rank.

68 Physical Measurement Equations

68.1 Time

The time bridge predicts $\Delta t_k = T_{\text{clock}}(\Xi_k)$. A calibrated clock yields Δt_k^{obs} with uncertainty σ_t . A basic residual is $r_t = (\Delta t_{\text{pred}} - \Delta t_{\text{obs}}) / \sigma_t$. The bridge is empirically meaningful only if T_{clock} is fixed before the holdout interval.

68.2 Energy and spectroscopy

A quantum bridge supplies an energy-valued Hamiltonian. For an emission from $E_a > E_b$ use $\Delta E_{ab} = E_a - E_b$ and $\nu_{ab} = \Delta E_{ab} / h$; an unordered pair has positive line frequency $|E_a - E_b| / h$. Line centers and systematic uncertainties require independent calibration.

68.3 Geometry

A spacetime bridge supplies $g_{\mu\nu}(x; \Xi)$. Proper time, invariant distance, redshift, lensing, or tidal observables must be calculated from that metric and compared with clock/interferometric/astronomical data. Internal G cannot be substituted directly because it has positive signature and different type.

68.4 Quantum outcomes

Given independently prepared Ξ , bridge $\rho_Q(\Xi)$, and calibrated POVM effect E_a , $p(a|\Xi) = \text{tr}(E_a \rho_Q(\Xi))$. Repeated outcomes define a binomial/multinomial likelihood. This gives an actual statistical test of the bridge.

68.5 Particle observables

Π_{SM} must output masses, charges, couplings, mixings, and decay/scattering parameters. Comparison targets include PDG values and cross sections. A group-theoretic resemblance without numerical predictions is not sufficient.

68.6 Biological observables

A biological bridge must define boundaries and species before J_{in} , J_{out} and J_{net} can be measured. Viability predictions require a time horizon and perturbation class, not only an instantaneous flux balance.

68.7 Subject-level observables

World/Self variables require task-dependent operational readouts and interventions. Γ_{subj} must predict independent behavioral/physiological/report data before any phenomenal interpretation is considered.

69 Falsification Protocol for the World Equation

69.1 Freeze the mathematical kernel

No equation in the finite kernel is changed after observing the physical holdout data.

69.2 Freeze the bridge

The map from Ξ to the tested observable, all calibration constants, nuisance parameters, and preprocessing are specified on training/calibration data.

69.3 Independent preparation

The relevant Ξ components are identified without using the same outcome that will test the prediction.

69.4 Predict

The BFG produces a point prediction or probability distribution with uncertainty.

69.5 Compare

A pre-specified statistic compares prediction and holdout data against strong baselines using the same inputs.

69.6 Interpret failure locally

A failed quantum bridge falsifies that quantum identification first; it does not logically falsify the matrix identities of the finite kernel. Repeated failures across bridge classes can, however, undermine the claim that the BFG describes physical reality.

70 Dimensional-Consistency Handbook

70.1 Dimensionless kernel sector

The canonical finite kernel is most naturally formulated with Y, P, Q, C, B, G , the partial isometries, projectors, SVD frames, and witness density operators dimensionless. This is consistent because these objects encode geometry, support, relative weighting, or normalized state rather than an SI-valued physical observable. Singular values of the dimensionless analysis operator are then dimensionless and so is $Y_+ = \Sigma^2$.

70.2 Formation sector

K and ρ_F are assigned the common formation unit E_F in the canonical mathematical class. This choice is not a claim that E_F is joules. It is a type constraint ensuring that $K_+ = U^\dagger (K \oplus K) U$ and $\rho_{F,+}$ can be compared inside the seed construction. If a physical bridge later identifies K with an energy-like operator while ρ_F is assigned a different physical dimension, the constant κ_F performs the one-time unit conversion in $\rho_{\text{seed}} = -\kappa_F \lambda \Pi_{\text{seed}}$.

70.3 Witness sector

ρ_w is normalized and dimensionless. Projector traces against ρ_w are therefore dimensionless probabilities inside the kernel. The product $m_{\text{tot}} = m_{\text{sel}} m_4$ is a conditional survival probability for the two-stage mathematical selection. Calling it a physical detection probability requires a separate instrument bridge.

70.4 Time sector

The event index k has no time dimension. ε in the scalar normal form can be dimensionless if both equations use the same model-time parameter, or it can encode a ratio of time scales after nondimensionalization. Physical seconds enter only through T_{clock} . This prevents the common mistake of assigning SI seconds to a recursion count without calibration.

70.5 Spacetime sector

Use $x^0 = ct$ and all x^μ in metres when g is dimensionless. SI proper time is $d\tau = c^{-1} \sqrt{-g_{\mu\nu} dx^\mu dx^\nu}$ for a timelike displacement. Natural units $c=1$ must be converted before reporting seconds. Coordinates mixing seconds and metres cannot have all metric components dimensionless. In the natural-unit disformal convention $[\varphi] = M$, $[\partial \varphi] = M^2$ and $[\lambda] = M^{-4}$.

70.6 Quantum and particle sector

ρ_Q and POVM effects are dimensionless. H_{hat} carries energy. Mass can be quoted in kg or energy units divided by c^2 ; cross sections carry area; decay widths carry energy and lifetimes time. A successful Π_{SM} must

therefore output quantities with these physical dimensions or fixed conversion constants. Dimensional matching is necessary but not sufficient for a correct bridge.

71 Stratum and Regime Manual

71.1 Selection-gap stratum

The selected spectral subspace remains fixed in dimension as long as no eigenvalue of Y crosses the threshold 1. Inside such a region Q varies smoothly as an invariant spectral projector. At a crossing, the selected rank changes. The event is not an infinitesimal perturbation inside the old stratum; it is a transition to a new active model stratum.

71.2 Polar-rank stratum

For X_{sel} or X_{R4} , constant rank means the number of positive singular values remains fixed. The polar partial isometry and its source/target projectors then admit the constant-rank derivative used in this paper. When the smallest positive singular value reaches zero, the pseudoinverse loses a uniform bound and the support dimension changes.

71.3 Dual-load stratum

The formulas for α and β require $\Lambda_C + \Lambda_B > 0$. The canonical dual-load analysis further assumes both loads positive. If one load vanishes, a one-channel limiting rule would constitute a different model branch unless explicitly included. The present canonical branch therefore treats failure of the required dual-load condition by its declared terminal semantics.

71.4 Simple-SVD stratum

The componentwise SVD derivative assumes positive simple singular values. At exact degeneracy, the singular subspace can remain well defined while individual vectors are gauge-dependent. The quotient state remains meaningful; the simple-frame differential is replaced by a block/subspace description if one wishes to cross the degeneracy smoothly.

71.5 Seed stratum

A simple strictly negative lowest complement eigenvalue defines the rank-one seed. Its sign and eigenvalue gap must remain separated on a regular seed stratum. If $E \neq 0$ with $\lambda_{\min} \geq 0$ the event is terminal; a multiple negative minimum is also terminal. The no-seed branch is $E=0$. Terminal surfaces do not inherit a nonterminal differentiability claim.

72 Proof Appendix Core Invariants

72.1 Positive Gram invariant

Σ is a diagonal matrix of positive singular values in the reduced SVD. For any vector x , $x^\dagger Y_- x = x^\dagger \Sigma^2 x = \|\Sigma x\|^2 \geq 0$. Hence Y_- is positive semidefinite. Adding the identity gives $x^\dagger G_- x = \|x\|^2 + \|\Sigma x\|^2 > 0$ for nonzero x .

The positive dual-load ambient analysis strengthens this positivity result to $0 < Y_- < I$ by the bound proved in Section 17. The result is not monotone comparison with the old Y matrix, since the carrier and coordinates can change.

72.2 Hermiticity of K_-

If $K = K^\dagger$ then $K \oplus K$ is Hermitian. Therefore $K_-^\dagger = [U^\dagger (K \oplus K) U]^\dagger = U^\dagger (K \oplus K)^\dagger U = K_-$. This guarantees a real seed spectrum after compression.

72.3 Positivity under transport

For any positive ρ and arbitrary T , $x^\dagger T \rho T^\dagger x = (T^\dagger x)^\dagger \rho (T^\dagger x) \geq 0$. Thus both selected and R4-transported formation/witness numerators remain positive.

72.4 Witness normalization

Cyclicity of the trace gives $\text{tr}(T_4 \rho_W^{\text{sel}} T_4^\dagger) = \text{tr}(T_4^\dagger T_4 \rho_W^{\text{sel}}) = \text{tr}(S_4^\wedge \rho_W^{\text{sel}}) = m_4$. Division by positive m_4 gives trace one.

72.5 Projector successor

$S_4^\wedge +$ and Π_{seed} are self-adjoint idempotents and $\Pi_{\text{seed}} \leq E = I - S_4^\wedge +$. Therefore $S_4^\wedge + \Pi_{\text{seed}} = \Pi_{\text{seed}} S_4^\wedge + = 0$. Expanding P_{-+}^2 leaves exactly $S_4^{\wedge++} \Pi_{\text{seed}}$.

72.6 Gauge covariance

Under an allowed target unitary W , every target operator O transforms to $W^\dagger O W$. Products, sums, spectra, traces, positivity, rank, and projector relations are preserved under unitary conjugation. Therefore the equivalence class, rather than any chosen matrix basis, is the canonical successor.

72.7 Survival bounds

For a density operator ρ and projector S , $0 \leq S \leq I$. Positivity gives $\text{tr}(S\rho) \geq 0$, while $\text{tr}[(I-S)\rho] \geq 0$ gives $\text{tr}(S\rho) \leq \text{tr}(\rho) = 1$. Apply this to both conditioned stages to obtain m_{sel}, m_4 in $[0,1]$.

For this kernel the second bound sharpens to $m_4 = 1$. Carrier dimension and mass must still be distinguished: the seed may enlarge P_{-+} within C^r while the witness remains on $S_4^{\wedge+}$.

73 Measurement Protocol Time and Geometry

73.1 Clock calibration

Select a physical preparation family and independently measured clock intervals. Fit or derive T_{clock} only on calibration data. Freeze the map. For each holdout BFG state, predict the next interval and compare with traceable SI clock measurements. A systematic state-dependent residual indicates failure of the clock bridge even if the event ordering remains correct.

73.2 Metric calibration

A spacetime bridge must specify coordinates or coordinate-invariant predictions, the relation between recursive state and matter/source variables, and any constants. The preferred tests are invariant or operational: proper-time ratios, frequency redshift, interferometric phase, geodesic deviation, lensing, or gravitational-wave propagation. Coordinate components alone are not observables.

73.3 Separating internal and spacetime metrics

No fit is allowed to set $g_{\mu\nu}$ numerically equal to the positive G matrix merely because both are called metrics. Their dimensions, signatures, domains, and transformation laws differ. A valid bridge is a typed map, not a relabeling.

73.4 Gravitational falsification

Once the bridge and source model are frozen, predictions can be compared with a standard gravitational baseline. A failure means that the chosen geometry/dynamics bridge is rejected. If every viable bridge requires post-hoc modification, the stronger claim that the BFG explains gravity loses empirical content.

74 Measurement Protocol Quantum and Particle Physics

74.1 State preparation

A preparation protocol must determine the BFG state or the components needed by Π_Q without using the later measurement outcome. Tomographic data used to define ρ_Q cannot be reused as independent confirmation of the same bridge.

74.2 Instrument specification

The POVM or instrument is calibrated independently. For a fixed effect E_a , the bridge predicts $p_a = \text{tr}(E_a \rho_Q)$. Repetition produces a likelihood. Competing BFG bridges are compared on the same preparations and effects.

74.3 R4 compatibility test

If one proposes that an actual success Kraus operator is T_4 , then its effect must equal $S_4^{\wedge} = T_4^\dagger T_4$ and the physical output transport must match the R4 target coordinates. General nonprojective effects require additional channels and cannot be identified directly with a partial isometry.

74.4 Particle parameter test

A Π_{SM} bridge must output a finite vector of quantities such as masses, charge assignments, mixing parameters, couplings at a specified renormalization scale, widths, and branching fractions. These predictions are compared with independent PDG values and collider/precision measurements. Merely reproducing $SU(3) \times SU(2) \times U(1)$ notation is not a numerical test.

74.5 Holdout principle

Calibration constants may be estimated on a declared training set. The test set must remain untouched until predictions are frozen. The same baselines and uncertainty treatment are applied to BFG and comparison models.

75 Measurement Protocol Atomic, Molecular, and Biological Sectors

75.1 Atomic closure

A quantitative atom bridge must specify constituent degrees of freedom, Hamiltonian or open-system generator, charge/mass inputs, and the map from recursive state to these quantities. Spectral lines, ionization energies, transition rates, and scattering observables provide direct tests.

75.2 Molecular closure

A molecular bridge must define nuclear/electronic or effective degrees of freedom and predict binding energies, equilibrium geometries, vibrational spectra, or reaction observables. The symbolic Cl_{NM} becomes a real mathematical operator only when its input/output spaces and update are specified.

75.3 Cellular closure

For a cell-level bridge, define the physical boundary, transported species, energy carriers, internal formation variables, and repair variables. Measure gross influx and outflux separately. Viability is tested over a horizon under a specified perturbation class, not by one stationary snapshot.

75.4 Causal minimality

A claim $n_{\min}^{\text{causal}}=3$ requires an admissible three-role realization and exclusion of equivalent one- and two-role realizations inside the same declared equivalence class. Four-or-more-role realizations need not be excluded because they may simply be nonminimal.

75.5 Multiscale closure

Higher biological levels require a coarse-graining map and a test that the proposed macrostate captures future-relevant dynamics. Information synergy means that variables cannot be discarded merely because each is individually weak. Intervention tests are required for causal claims.

76 Measurement Protocol World, Self, and Experience

76.1 World readout

A world variable W should predict externally oriented observations beyond what a chosen reduced baseline already predicts. Its definition must specify inputs, history dependence, horizon, and loss function.

76.2 Self readout

A self variable S should predict or causally regulate the system's own future state/action. Interventions on S must be defined within the admissible model class; destructive interventions do not establish the intended causal role.

76.3 World/Self closure

A closed U_{WS} must reproduce the projected successor of the underlying BFG dynamics. If two underlying states share the same (W,S) but produce different projected successors, the proposed functional state is not closed.

76.4 Witness integration

The kernel witness must be linked to the functional World/Self variables by an explicit map. Equality of normalized matrices or norms is not a lineage criterion. Directed transport and unnormalized survival bookkeeping remain separate.

76.5 Phenomenal hypothesis

Only after functional criteria are operational and independently measured can H_{phen} be tested against report, behavior, physiology, perturbation, or other declared evidence. No purely mathematical definition can by itself supply empirical confirmation of phenomenality.

77 Plain-Language Glossary for Readers

77.1 Formation

Formation is the positive operator ρ_F that carries the magnitude and structure of what is being recursively retained or newly seeded. It is intentionally not normalized like a probability distribution.

77.2 Witness

Witness is the positive trace-one ρ_w . The first selection can remove original witness mass. The present ambient R4 then transports its selected support isometrically, so $m_4=1$. Historical mass is recorded separately from the five-component normalized state.

77.3 Gram geometry

Y and G describe an internal positive geometry used to select and measure active recursive directions. This geometry is endogenous because it is both an input to and an output of the update.

77.4 Selection

Selection means restriction to the subspace whose Y-spectrum lies in the declared interval and overlaps the current persistent carrier. Polar transport preserves the directed relation between source and target supports.

77.5 Load

A load is a scalar obtained by evaluating transported formation through one neutral/balance channel in the current internal metric. Loads determine the relative weights of the analysis branches.

77.6 R4

R4 is the second recursive re-alignment step. It compares the SVD source frame with the selected carrier and produces the partial isometry used for final witness/formation transport.

77.7 Seed

A seed is positive formation from a simple negative lowest complement mode. When $E=0$ no seed is required. When $E \neq 0$ without such a unique negative mode, this kernel terminates.

77.8 Admissibility

Admissibility is the conjunction of the gates required for a particular model class. It is not identical to stability or viability; a state can satisfy a static gate and still leave the admissible region on the next update.

77.9 Persistence

Persistence means continued admissible directed evolution of the relevant typed entity. Witness persistence, formation persistence, biological viability, and subject persistence are different typed predicates.

77.10 Bridge law

A bridge law maps the mathematical BFG state to a physical or functional observable. Bridge laws are where the world equation becomes experimentally vulnerable.

78 Equation Roadmap Reading One Complete World-Equation Step

78.1 Input

Begin with the declared five-component finite state Ξ_k . This is the state chosen for the model; no absolute minimality theorem is claimed. The historical witness record is additional trajectory bookkeeping.

78.2 Neutral decomposition

Compute $C_k = (I + Y_k)^{-1}$ and $B_k = I - C_k$. These operators divide the current Gram load into complementary channels without introducing an external parameter.

78.3 Spectral admissibility

Compute $Q_k = 1_{[0,1)}(Y_k)$. This selects the currently sub-threshold spectral sector. Intersect it with persistence through $X_{\text{sel}} = Q_k P_k$.

78.4 Directed selection transport

Take $T_{\text{sel}} = \text{polar}(X_{\text{sel}})$. Read $S_{\text{sel}}^{\wedge}$ from $T_{\text{sel}}^{\dagger} T_{\text{sel}}$ and P_{sel} from $T_{\text{sel}} T_{\text{sel}}^{\dagger}$. These projectors are the actual source and target supports of the first directed transport.

78.5 First survival bookkeeping

Transport formation and witness. Compute m_{sel} before normalizing the witness. If $m_{\text{sel}} = 0$, terminate. Otherwise condition ρ_w^{sel} while retaining m_{sel} as historical survival information.

78.6 Endogenous metric

Compute $G = I + Y$ and the G -orthogonal projector P_G onto the selected target support. The current geometry now directly controls how formation is evaluated.

78.7 Endogenous loading

Evaluate Λ_C and Λ_B and construct α, β . No free branch weight is introduced on the dual-load stratum.

78.8 Recursive analysis

Construct A from the weighted C/B branches. Its reduced SVD supplies the active rank r , the next dimension, the successor Gram spectrum, and the frames required by K_+ and R_4 .

78.9 Successor geometry and kernel

Set $Y_+ = \Sigma^{\wedge 2}$ and $K_+ = U^{\dagger}(K \oplus K)U$. At this point the next active space is already fixed as $C^{\wedge r}$.

78.10 Recursive re-alignment

Form $X_{R_4} = V^{\dagger} P_{\text{sel}}$. The positive C -channel implies it is already a partial isometry, with source P_{sel} . It realigns the selected carrier and fixes the free target complement.

78.11 Second survival bookkeeping

The nominal second mass is identically $m_4 = 1$. Therefore $m_{\text{tot}} = m_{\text{sel}}$. Update the separate original-mass record $w_{k+1} = w_k m_{\text{sel},k}$ before interpreting a normalized witness as retention.

78.12 Formation inheritance

Transport selected formation by T_4 without normalization. This gives the inherited successor formation.

78.13 Complement test

Compute $E = I - S_4^+$. If $E=0$ there is no free target complement. If E is nonzero, compress K_+ to $\text{Ran}(E)$ and inspect its spectrum.

78.14 Seed or no-seed

If $E=0$ continue without seed. If $E \neq 0$ a simple strictly negative minimum produces the seed; otherwise terminate as T_{seednone} or T_{seeddeg} . No numerical tie-break replaces the rank-one rule.

78.15 Output

Assemble $\Xi_{k+1} = (K_+, \rho_F, \rho_W, Y_+, P_+)$ and then pass to its unitary equivalence class. The next event repeats the same algorithm on the newly generated active space.

79 Figure Companion What Each 3D Graphic Means

Figure 1 Section 3 Conditional readout. Conditional readout. $\phi_E = (E_{\text{exp}}/E_{\text{bind}})(I/I_0)$; ratios are independent inputs, not identified Xi data. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 2 Section 4 Computed matrix family. Computed matrix family. $Y = \text{diag}(y_1, y_2)$; $G = I + Y$; the positive metric is internal, not spacetime. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 3 Section 5 Computed projector. Computed projector. $Q = 1_{[0,1)}(Y)$; threshold 1 defines a discontinuous change of selected rank. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 4 Section 6 Computed control family. Computed control family. $Y = \text{diag}(1/2, 2)$, $P = vv^T$ and $W = \text{diag}(p, 1-p)$; polar selection has source P. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 5 Section 7 Computed ambient kernel. Computed ambient kernel. $Y = \text{diag}(y_1, 0.72)$; $P = F = W = vv^T$; $v = (\cos \theta, \sin \theta)$; formation unit $E_F = 1$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 6 Section 8 Computed ambient kernel. Computed ambient kernel. $A: H$ to H direct sum H ; no restriction to the selected rank-one carrier. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 7 Section 9 Computed ambient kernel. Computed ambient kernel. The second singular value can vanish on a rank boundary; an arbitrary full-SVD padding is excluded. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 8 Section 10 Computed ambient kernel. Computed ambient kernel. $Y = \text{diag}(y_1, y_2)$, $P = F = I$; coupled global formation loads determine alpha and beta. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 9 Section 11 Computed ambient kernel. Computed ambient kernel. $m_4 = 1$ to numerical precision; the exact proof uses $\text{Ran}(P_{\text{sel}})$ contained in $\text{Ran}(A^{\dagger})$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 10 Section 12 Computed bookkeeping. Computed bookkeeping. $w_{\text{next}} = w * m_{\text{sel}}$; conditioned witness trace remains one and cannot encode this product. Horizontal and vertical coordinate ranges are the sampled

grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 11 Section 13 Computed spectral control. Computed spectral control. A one-dimensional complement with $\lambda = -x$ gives seed trace κx ; each κ is a separate fixed-unit model. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 12 Section 14 Computed rule. Computed rule. For a nonempty complement, $\lambda < 0$ and a simple minimum are required; $\lambda \geq 0$ is terminal. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 13 Section 15 Computed bound. Computed bound. 3 times $\|\Delta X\|_2$ divided by $\sigma_{\min,+}$; an upper bound is not an attained response. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 14 Section 16 Conditional perturbation scale. Conditional perturbation scale. Illustrates inverse-gap scaling of the reduced resolvent; not a full derivative of an arbitrary kernel. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 15 Section 17 Exact scalar kernel. Exact scalar kernel. $P=W=1$ and $F>0$; $f(y)=2y^2/[(1+y)^2(1+y^2)]$; K and F remain neutral. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 16 Section 18 Conditional stability model. Conditional stability model. Assumes $V=z^2$ and $V_{\text{next}}=(1-r)V$; no such global BFG certificate is constructed here. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 17 Section 19 Computed mediator control. Computed mediator control. $N_B R N_B \dagger = n^2 R$; amplitude mediation uses $N_A = n^2$, not $N_A = n$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 18 Section 20 Schematic constraint model. Schematic constraint model. A margin diagram only; actual viability requires the unique nonterminal BFG trajectory and specified gates. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 19 Section 21 Computed conditional ode. Computed conditional ode. $a=1.5$, $\mu>0$, $b<1$; this positive branch exists because $a>1-b$. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 20 Section 22 Schematic countermodel. Schematic countermodel. Projection $P_i(x,h)=x$ with successor $x+h$ is not closed; this is a fiber test, not a BFG bridge. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 21 Section 23 Exact scalar kernel. Exact scalar kernel. One reclosure event is counted; no conversion to physical seconds is claimed. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 22 Section 24 Schematic embedding. Schematic embedding. An illustrative surface is not a Lorentz metric, gravitational field equation or prediction. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 23 Section 25 Assumed potential model. Assumed potential model. $V=x^2+y^2$ is a pedagogical potential; stability of spacetime must follow from a specified action and bridge. Horizontal and vertical

coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 24 Section 26 Conditional quantum model. Conditional quantum model. For a calibrated $E0=|0\rangle\langle 0|$, $p=\text{tr}(E0 \rho)$; the state-identification map is additional physics. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 25 Section 27 Computed bridge comparison. Computed bridge comparison. At (0.75,0.25), hypotheses A and B differ by 0.5; no actual measured outcomes are used. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 26 Section 28 Computed effect test. Computed effect test. A direct R4 effect is a projector. Generic positive effects with interior eigenvalues fail that test. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 27 Section 29 Schematic dependency. Schematic dependency. The surface labels a parameter dependency only; no Standard Model mass or coupling is derived. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 28 Section 30 Conventional conditional model. Conventional conditional model. $E(r)=1/(2r^2)-a/r$; ordinary quantum scaling, not a new BFG atomic spectrum. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 29 Section 31 Schematic binding landscape. Schematic binding landscape. The illustrative energy x^2+y^2-1 marks a negative binding region; it is not a BFG molecular potential. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 30 Section 31 Computed flux identity. Computed flux identity. $J_{\text{net}}=J_{\text{out}}-J_{\text{in}}$. Stationary net flux permits nonzero gross flows; no cell viability is inferred. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 31 Section 32 Schematic functional model. Schematic functional model. A product is only a visual aid. Π_W, Π_S , lineage and causal closure must be independently defined. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

Figure 32 Section 33 Schematic status panel. Schematic status panel. The zero sheet denotes absence of a derived prediction, not absence of subjective experience. Horizontal and vertical coordinate ranges are the sampled grids; the plotted quantity is identified on the height axis. Surface colour follows height and has no additional physical meaning.

80 Frequently Asked Technical Questions

80.1 Is the BFG one equation or many

It is one quotient evolution equation with an explicit factorization. The sub-operators are comparable to the factors of a numerical integrator or a structured map: they specify how the single successor is computed.

80.2 Why not use a scalar state

The five-component operator tuple is the declared state of this construction. It resolves spectra, magnitude, normalized witness and moving supports explicitly. This is not a proof that every alternative representation requires five components.

80.3 Why is the successor dimension allowed to change

Because the reduced SVD treats only active positive singular directions as successor coordinates. Rank is therefore a dynamical output. Fixed-dimensional padding would introduce null directions whose physical and seed status would be arbitrary.

80.4 Does gauge quotienting hide physical information

No information invariant under allowed coordinate changes is removed. Spectra, traces, positivity, rank, masses, and all simultaneously conjugated relations are preserved. The quotient removes only coordinate labels.

80.5 Is m_{tot} a Born probability

Not by definition. It is a mathematically normalized two-stage witness survival probability inside the kernel. A physical Born/instrument interpretation is a separate bridge hypothesis.

80.6 Is G spacetime

No. G is positive definite and internal. Spacetime is Lorentzian and requires a separate typed bridge.

80.7 Does scalar damping prove stability of the universe

No. It proves one scalar inequality on a declared branch. Full-state stability requires control of every endogenous derivative path and must account for neutral directions.

80.8 Does the Standard Model follow already

Not numerically. The BFG supplies a reconstruction program, but masses, charges, couplings, mixing parameters, and scattering observables must be generated by a fixed Π_{SM} and tested.

80.9 Does witness mean consciousness

No. Witness is a mathematical persistence role. World/Self and phenomenal experience require additional operational and empirical bridges.

80.10 What would make the BFG a physical world equation

A finite set of frozen measurement bridges that convert independently prepared BFG states into successful quantitative predictions across time, geometry, quantum/particle, and higher-level sectors without post-hoc remapping.

81 Canonical BFG at a Glance

$$[\Xi_{k+1}] = \text{Ubar}_{\text{BFG}} \setminus ([\Xi_k])$$

Selection $\rightarrow$ first transport $\rightarrow$ positive metric $\rightarrow$ endogenous dual loads $\rightarrow$ ambient reduced SVD $\rightarrow$ successor K and $Y \rightarrow$ isometric $R_4 \rightarrow$ formation inheritance $\rightarrow$ $E=0$ continuation or negative seed or terminal outcome $\rightarrow$ quotient successor.

Core invariants: Hermitian K ; positive formation; trace-one witness with separately recorded original mass; $0 < Y_+ < I$; orthogonal P ; $n_+ \leq n$. R_4 gives $m_4=1$. Nonterminal subsequent events preserve witness mass; seed termination remains possible.

Core non-universalities: positive Gram geometry is not Lorentz spacetime; witness survival is not automatically a Born probability; scalar damping is not full-state contraction; recursive event count is not physical seconds;

Cl_N labels are not transition laws until their operations are specified; functional subject structure is not by definition phenomenal experience.

Physical closure targets: T_{clock} for seconds; M_ϕ for the order observable; M_g for Lorentzian geometry and gravity; Π_Q plus calibrated instruments for quantum outcomes; Π_{SM} for masses, charges, couplings and mixing; atom/molecule/biology transition maps; Π_w , Π_s , U_{ws} and Γ_{subj} for functional subject structure.

Empirical rule: every bridge must be fixed before holdout testing. A successful prediction supports that bridge and its coupled kernel assumptions; a failed prediction rejects the tested bridge first. The Balance–Field Equation becomes a physical world equation only through a finite, frozen, independently testable measurement interface.

Completion status. Level-0 to Ξ , scalar information/order to Ξ , a closed Ξ -to- Ψ factor, calibrated physical time, Lorentzian dynamics, quantum instruments and particle parameters, quantitative atoms and molecules, biological viability and functional/phenomenal subject bridges are not constructed here. They remain explicit finite requirements, not consequences established by the illustrations.

Declarations

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Author contribution: Conceptualization, BFG synthesis, canonical finite-kernel formulation, interpretation, and manuscript responsibility: Marcel Theodor Wende. Co-authored precursor source lines are credited in References [1]–[3].

Data availability: No external empirical dataset is analyzed. The 3D computational panels use explicitly stated synthetic matrix families and independently checked scalar formulas. Open physical transitions are schematic or show conventional conditional models and are not BFG predictions of measured systems. Numerical controls support calculations, not proofs or empirical identification.

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Competing interests: No competing interest is asserted.

Ethics: No human participants, animal experiments, or identifiable personal data were used.

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